
DTU

Mathematical Modelling

Welcome to 02526

Welcome to 02526 – Mathematical Modelling

- Course responsibles:
  - Siavash Bigdeli
  - Anders Nymark Christensen
  - Dimitrios Papadopoulos

Welcome to 02526 – Mathematical Modelling

- Course responsibles:
  - ~~Siavash Bigdeli~~
  - ~~Anders Nymark Christensen~~
  - Dimitrios Papadopoulos (Course Responsible)
  - Mirza Karamehmedovic (Course co-responsible)

Team

- **Dimitrios:** Associate professor in section for Visual Computing
  – Current research: Computer vision, deep learning, AI



- **Mirza:** Associate professor in section for Scientific Computing
  – Current research: Inverse Problems, Scientific Computing, PDE



Teaching Assistants



Alessandro Covi



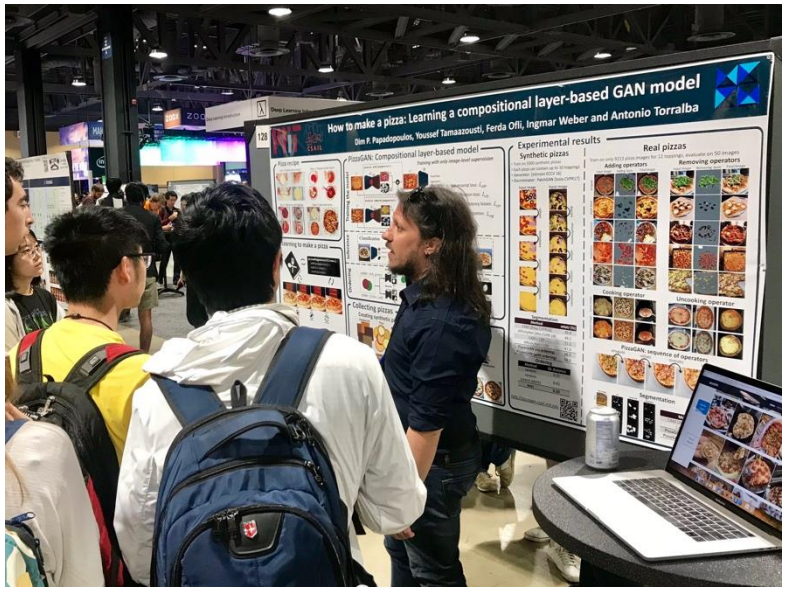
Victor Hannibal Folting Bjerre



Anders Niels Baunelin-Eriksen

Short Bio – Dimitrios Papadopoulos

- **Associate Professor** at the section of Visual Computing
- **Research interests:** Computer Vision and Machine Learning
- **Prior to DTU:**
  - Postdoc, MIT, MA, USA (2018 - 2021)
  - Visiting student, ETH Zurich, Switzerland (2016 - 2017)
  - PhD, University of Edinburgh, UK (2013 - 2017)
  - MSc and Meng at Democritus University of Thrace (2006-2012)



Today's Lecture

- Learning objectives
- Practicalities: Course Plan, Evaluation and exams
- Introduction to mathematical modeling
- Exercise 1: Salami Quality

What is the goal of the course?

Learning Objectives:

- Apply a mathematical model to solve a problem
- Explain the components of a mathematical model
- Judge area of validity of a mathematical model
- Use statistics to summarize the obtained results
- Implement numerical algorithms using Matlab or Python
- Set up test scenarios and collect data
- Explain the model parameter's influence on the model
- Interpret the calculated results of the model
- Visualize data and results graphically
- Evaluate the relevance of the results for the analyzed problem
- Write a stringent and structured report that precisely describes the model and analysis of the calculated results

These are valuable tools and can add value to your work

Learning objectives are quite generic

Working with models, reflecting upon relevance, validity, explanatory power …

Data visualization, programming and implementation

Improving and reflecting upon report writing

Course Plan

Course Plan

- The course is fully physical
- 13 weeks, Tuesday from 08.00 – 12.00
- 5 ECTS work load is ~9 hours per week (including the 4 hours each Tuesday)

Course Plan

- The course is fully physical
- 13 weeks, Tuesday from 08.00 – 12.00
- 5 ECTS work load is ~9 hours per week (including the 4 hours each Tuesday)

Date	What	Who
Tuesday 3/2	Welcome + Salami introduction	Dimitrios, Mirza
Tuesday 10/2	LDA + Scientific writing	Dimitrios
Tuesday 17/2	Peergrade and feedback	Dimitrios
Sunday 22/2	HAND IN - Salami	
Tuesday 24/2	Optical flow intro	Mirza
Sunday 1/3	Deadline - FeedbackFruit	
Tuesday 3/3	Optical flow	Mirza
Tuesday 10/3	Optical flow + Salami feedback	Mirza, Dimitrios
Sunday 15/3	HAND IN - Optical flow	
Tuesday 17/3	Integer programming	Stefan, Dimitrios
Sunday 22/3	Deadline - FeedbackFruit	
Tuesday 24/3	Exercise followup	Mirza
Tuesday 31/3	Easter break	
Sunday 5/4	HAND IN - Integer programming	
Tuesday 7/4	Optical flow feedback	Mirza
Tuesday 14/4	Exam project Q&A, Integer Programming feedback	All
Sunday 19/4	Deadline - FeedbackFruit	
Tuesday 21/4	Exam project	
Tuesday 28/4	Exam project	
Tuesday 5/5	Exam project	
Sunday 10/5	HAND IN - Exam project	
Thursday 21/5	Oral Exams	

Course Plan

- The course is fully physical
- 13 weeks, Tuesday from 08.00 – 12.00
- 5 ECTS work load is ~9 hours per week (including the 4 hours each Tuesday)
- **3 Exercises projects run over 3 weeks**
- **Hand in of reports plus peergrading**
- **1 Exam project an report over 4 weeks**

Date	What	Who
Tuesday 3/2	Welcome + Salami introduction	Dimitrios, Mirza
Tuesday 10/2	LDA + Scientific writing	Dimitrios
Tuesday 17/2	Peergrade and feedback	Dimitrios
Sunday 22/2	HAND IN - Salami	
Tuesday 24/2	Optical flow intro	Mirza
Sunday 1/3	Deadline - FeedbackFruit	
Tuesday 3/3	Optical flow	Mirza
Tuesday 10/3	Optical flow + Salami feedback	Mirza, Dimitrios
Sunday 15/3	HAND IN - Optical flow	
Tuesday 17/3	Integer programming	Stefan, Dimitrios
Sunday 22/3	Deadline - FeedbackFruit	
Tuesday 24/3	Exercise followup	Mirza
Tuesday 31/3	Easter break	
Sunday 5/4	HAND IN - Integer programming	
Tuesday 7/4	Optical flow feedback	Mirza
Tuesday 14/4	Exam project Q&A, Integer Programming feedback	All
Sunday 19/4	Deadline - FeedbackFruit	
Tuesday 21/4	Exam project	
Tuesday 28/4	Exam project	
Tuesday 5/5	Exam project	
Sunday 10/5	HAND IN - Exam project	
Thursday 21/5	Oral Exams	

Course Plan

- The course is fully physical
- 13 weeks, Tuesday from 08.00 – 12.00
- 5 ECTS work load is ~9 hours per week (including the 4 hours each Tuesday)
- 3 Exercises projects run over 3 weeks
- Hand in of reports plus peergrading
- 1 Exam project an report over 4 weeks
- **The work is done in groups of 3-4 people. Pairs can be acceptable, but solo is not recommended**
- **Report hand-in on Sunday before the next exercise**

Date	What	Who
Tuesday 3/2	Welcome + Salami introduction	Dimitrios, Mirza
Tuesday 10/2	LDA + Scientific writing	Dimitrios
Tuesday 17/2	Peergrade and feedback	Dimitrios
Sunday 22/2	HAND IN - Salami	
Tuesday 24/2	Optical flow intro	Mirza
Sunday 1/3	Deadline - FeedbackFruit	
Tuesday 3/3	Optical flow	Mirza
Tuesday 10/3	Optical flow + Salami feedback	Mirza, Dimitrios
Sunday 15/3	HAND IN - Optical flow	
Tuesday 17/3	Integer programming	Stefan, Dimitrios
Sunday 22/3	Deadline - FeedbackFruit	
Tuesday 24/3	Exercise followup	Mirza
Tuesday 31/3	Easter break	
Sunday 5/4	HAND IN - Integer programming	
Tuesday 7/4	Optical flow feedback	Mirza
Tuesday 14/4	Exam project Q&A, Integer Programming feedback	All
Sunday 19/4	Deadline - FeedbackFruit	
Tuesday 21/4	Exam project	
Tuesday 28/4	Exam project	
Tuesday 5/5	Exam project	
Sunday 10/5	HAND IN - Exam project	
Thursday 21/5	Oral Exams	

Course Plan

- The course is fully physical
- 13 weeks, Tuesday from 08.00 – 12.00
- 5 ECTS work load is ~9 hours per week (including the 4 hours each Tuesday)
- 3 Exercises projects run over 3 weeks
- Hand in of reports plus peergrading
- 1 Exam project an report over 4 weeks
- The work is done in groups of 3-4 people. Pairs can be acceptable, but solo is not recommended
- Report hand-in on Sunday before the next exercise
- **Introduction and follow-up lecture for each project**
- **You will read each other's report(s) and comment on them through Peergrade**
  - **Peergrading is mandatory, and a hand in will count for nothing if not done**

Date	What	Who
Tuesday 3/2	Welcome + Salami introduction	Dimitrios, Mirza
Tuesday 10/2	LDA + Scientific writing	Dimitrios
Tuesday 17/2	Peergrade and feedback	Dimitrios
Sunday 22/2	HAND IN - Salami	
Tuesday 24/2	Optical flow intro	Mirza
Sunday 1/3	Deadline - FeedbackFruit	
Tuesday 3/3	Optical flow	Mirza
Tuesday 10/3	Optical flow + Salami feedback	Mirza, Dimitrios
Sunday 15/3	HAND IN - Optical flow	
Tuesday 17/3	Integer programming	Stefan, Dimitrios
Sunday 22/3	Deadline - FeedbackFruit	
Tuesday 24/3	Exercise followup	Mirza
Tuesday 31/3	Easter break	
Sunday 5/4	HAND IN - Integer programming	
Tuesday 7/4	Optical flow feedback	Mirza
Tuesday 14/4	Exam project Q&A, Integer Programming feedback	All
Sunday 19/4	Deadline - FeedbackFruit	
Tuesday 21/4	Exam project	
Tuesday 28/4	Exam project	
Tuesday 5/5	Exam project	
Sunday 10/5	HAND IN - Exam project	
Thursday 21/5	Oral Exams	

Evaluation and exam

Exam

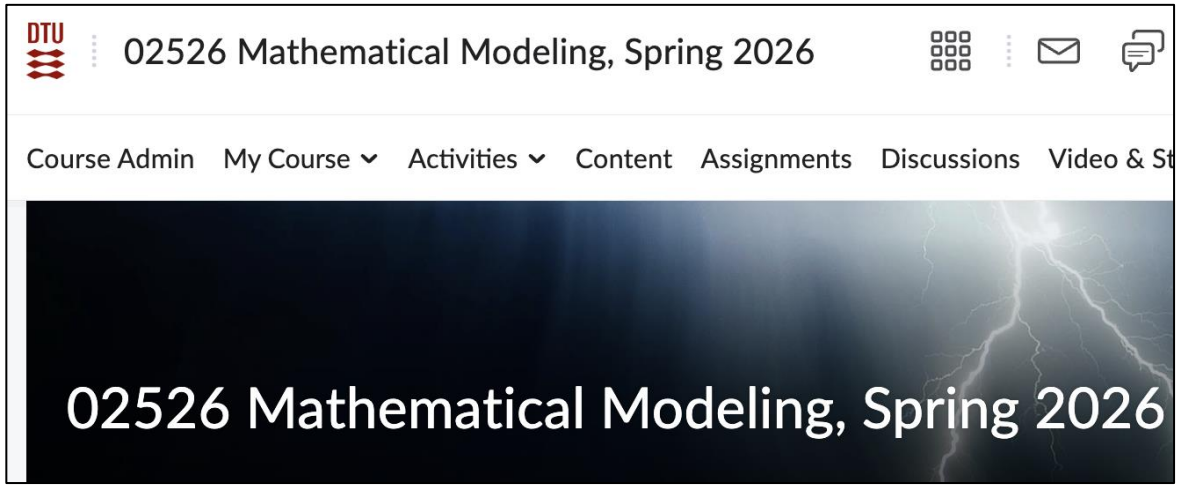
- The exercise reports (and peergrading) are pass/no pass.
  - Formative feedback given
  - *Must pass all exercise projects to pass the course*

- Exam project run over 4 weeks
  - Largely unassisted

- Short oral defense with the basis in the project report
  - Summative feedback
  - External examiner
  - Exam dates are yet to be settled

Useful Information, platforms and communication

DTU Learn

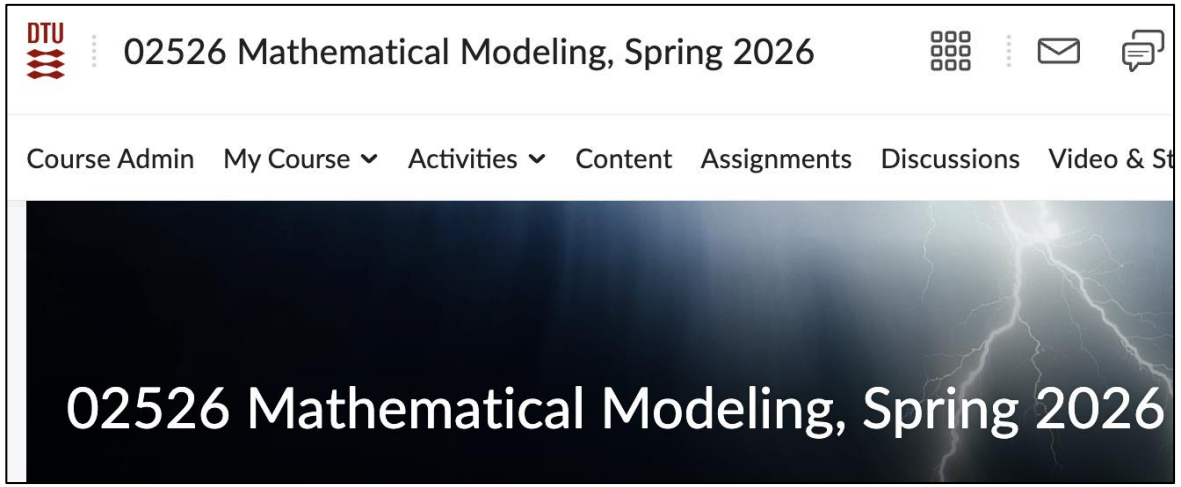


https://learn.inside.dtu.dk/d2l/home/296292

- All material (or links)
- Curriculum
- Lectures
- Exercises
- Handins/ Peergrade
- Announcements

Useful Information, platforms and communication

DTU Learn



https://learn.inside.dtu.dk/d2l/home/296292

- All material (or links)
- Curriculum
- Lectures
- Exercises
- Handins/ Peergrade
- Announcements

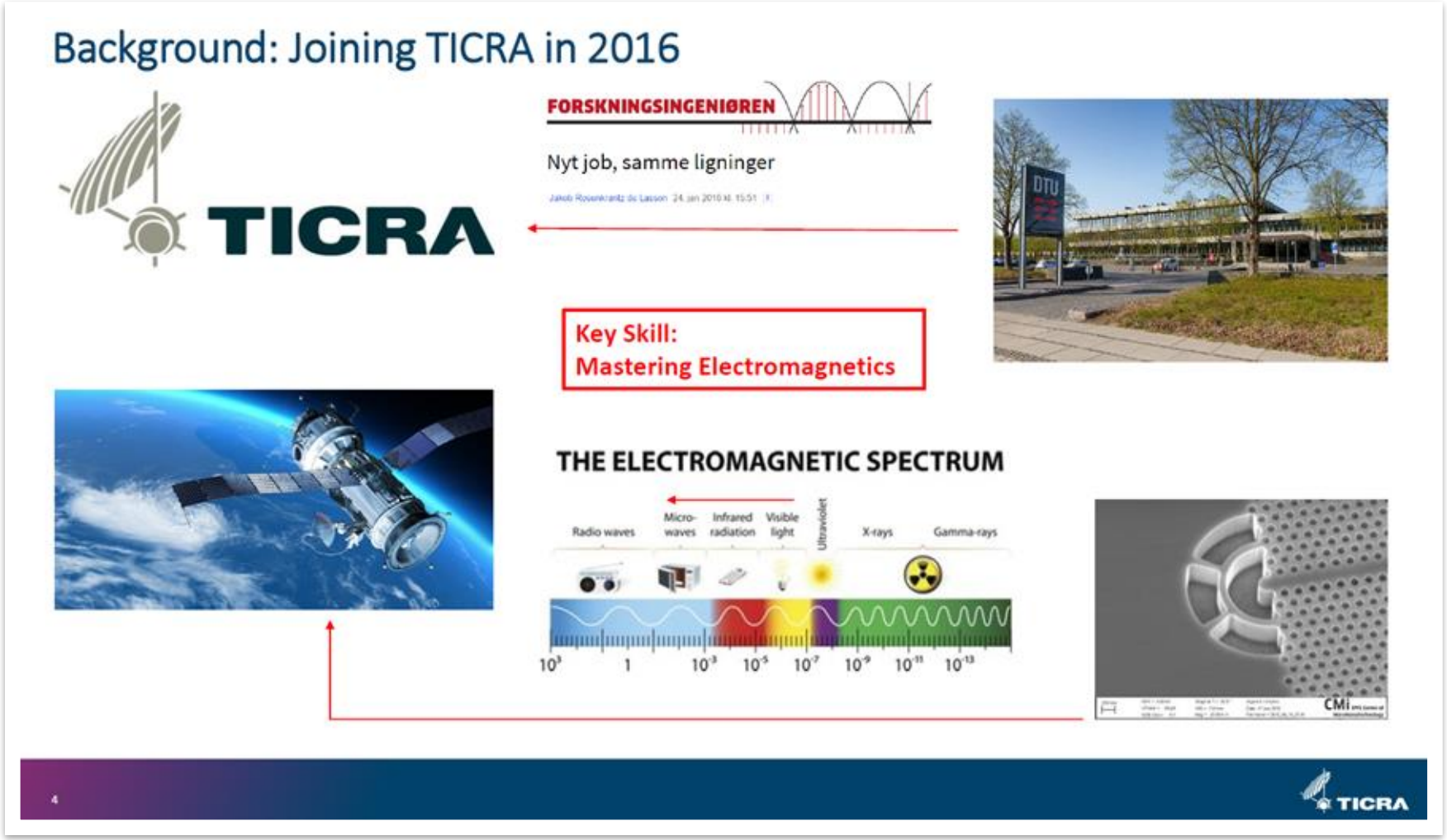
Slack workspace

https://join.slack.com/t/02526mathemat-0wh5248/shared_invite/zt-3pferlik0-uiJ0PE37ds1sUwZv2bstqQ

- Group members
- Questions for exercises
- General questions
- Announcements

What constitutes a good (physics) engineer

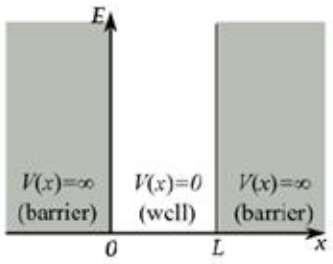
- Seminar here at DTU for the physics line, see https://ing.dk/blog/gode-fysik-ingenioer-223633 by Jakob Rosenkrantz de Lasson

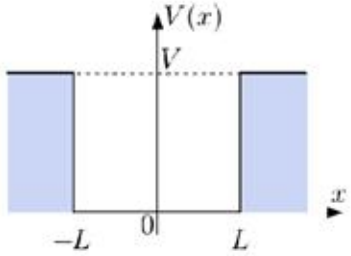


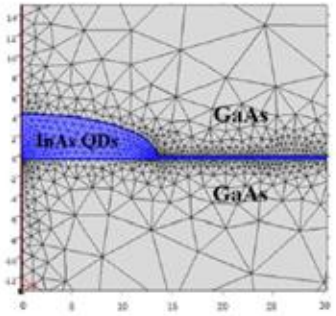
Mathematical Models

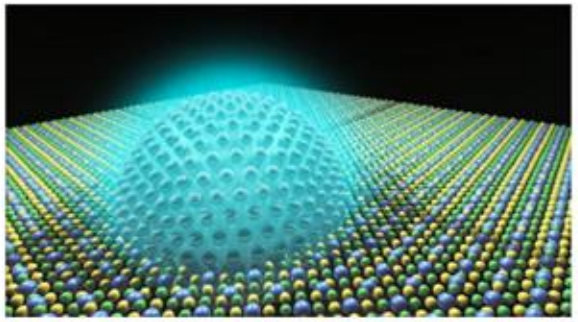
Ideal, simplified → Realistic, complex

Pen and paper → Computer simulation, lab

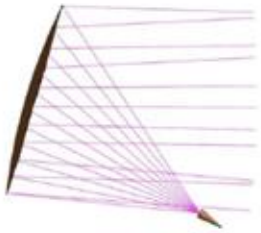


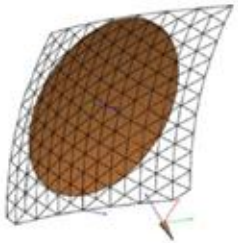






1. Analyze and understand idealized and simplified system → 2. Extend system/add effects/improve analysis method, and re-analyze







Key Skill:
Formulate, Understand and Analyze Mathematical Models

You'll Never Stop Writing



Software documentation



Design of Axially Corrugated Horn for Ka-Band Offset Reflector Antenna

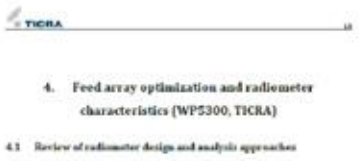
Technical Note 1

Provided by: TICRA, Denmark

Design/Consulting reports



R&D project proposals and reports



Conference papers



(...Some of it even in LaTeX 😮 *)*

Key Skill: Concise Writing

Input/Output and Data Processing

Field Work



Lab Work



Office Work

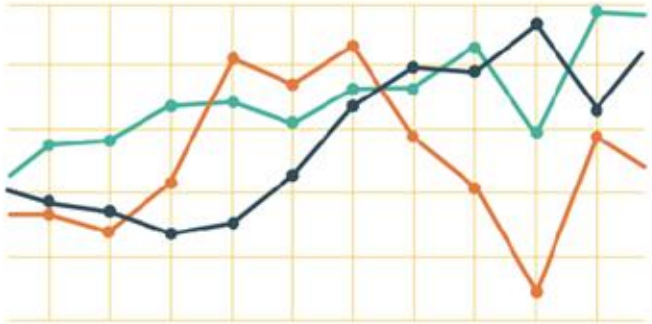


Key Skill:
Scripting (Matlab, Python, etc.)

Data Processing



Visualization



Input Files



Mathematical modeling

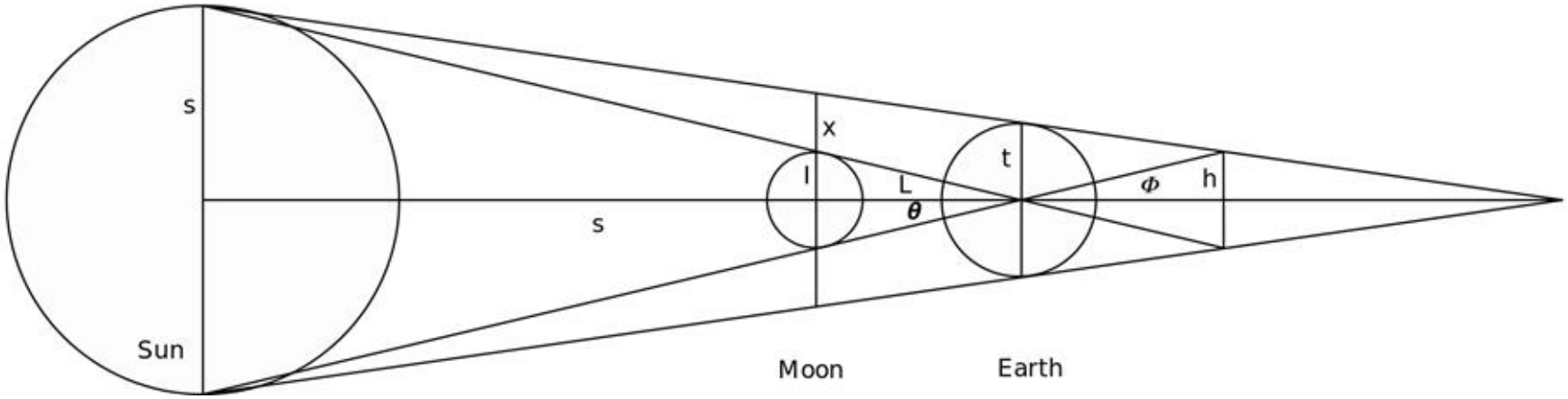
What is mathematical modeling?

A **mathematical model** is a **purposeful simplification** of **reality**.

- **Models are a way for us to describe the real world in terms of math, rules, relations etc.**
- **Allow us to make predictions and gain insights.**

The first mathematical models?

- Hipparchus 190-120 BC
  - Distance to the moon and sun
  - Size of the moon and sun

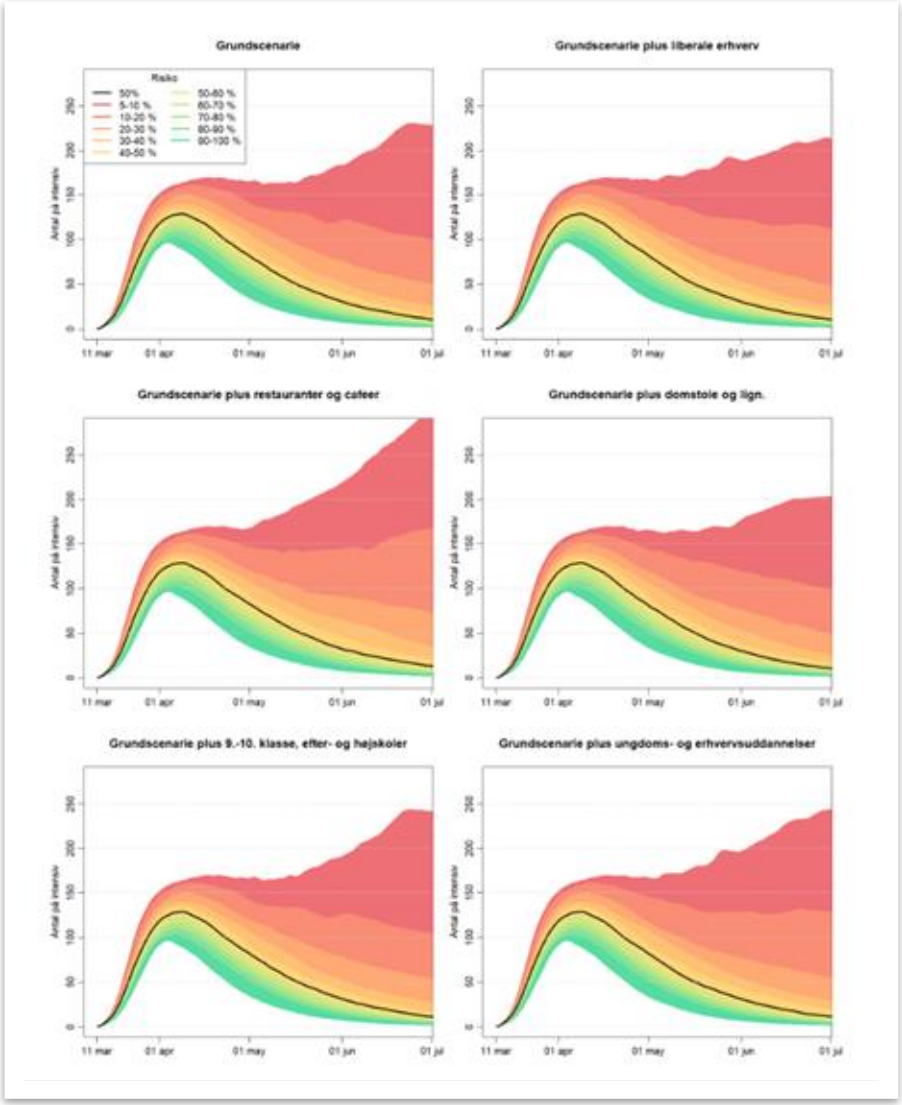


https://commons.wikimedia.org/wiki/File:HipparchusConstruction.svg

Mathematical modelling: During COVID



Photo: www.regeringen.dk

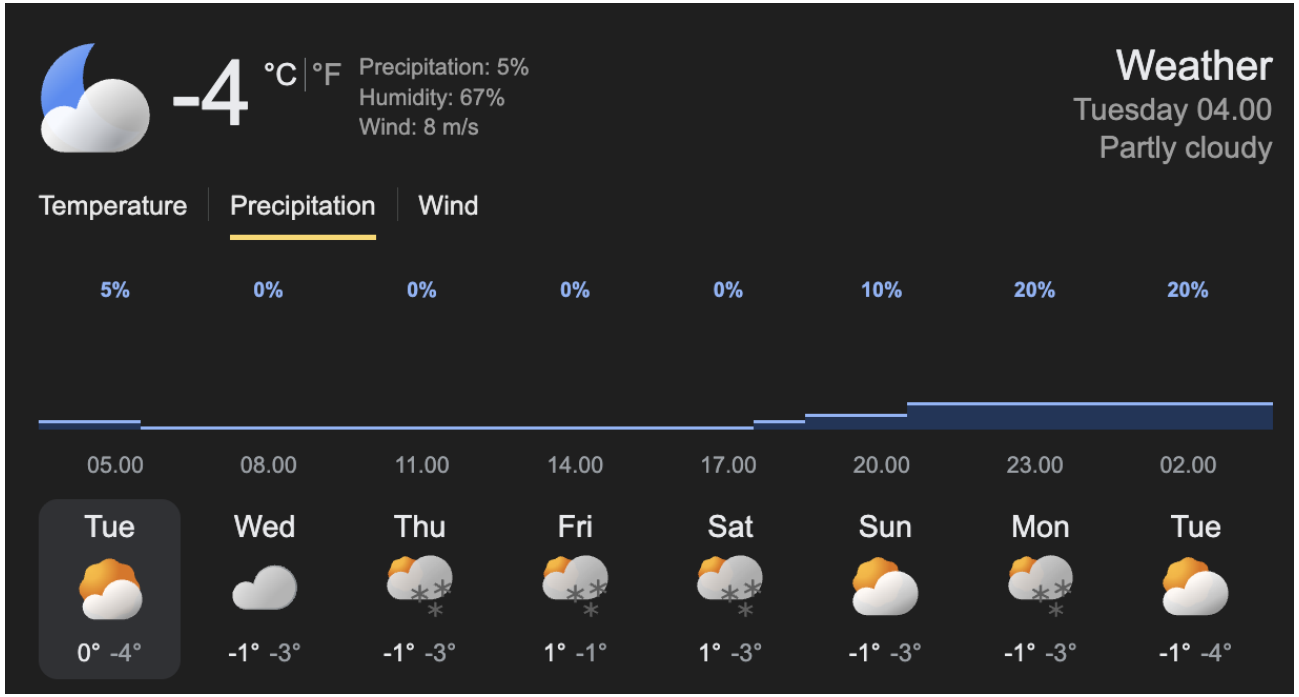


Source: SSI. ”Ekspertrapport: Matematisk modellering af COVID-19 smittespredning og sygehusbelastning ved scenarier for yderligere initiativer i Fase 1 af genåbningen”

Mathematical Modeling EVERYWHERE

Mathematical Modeling EVERYWHERE

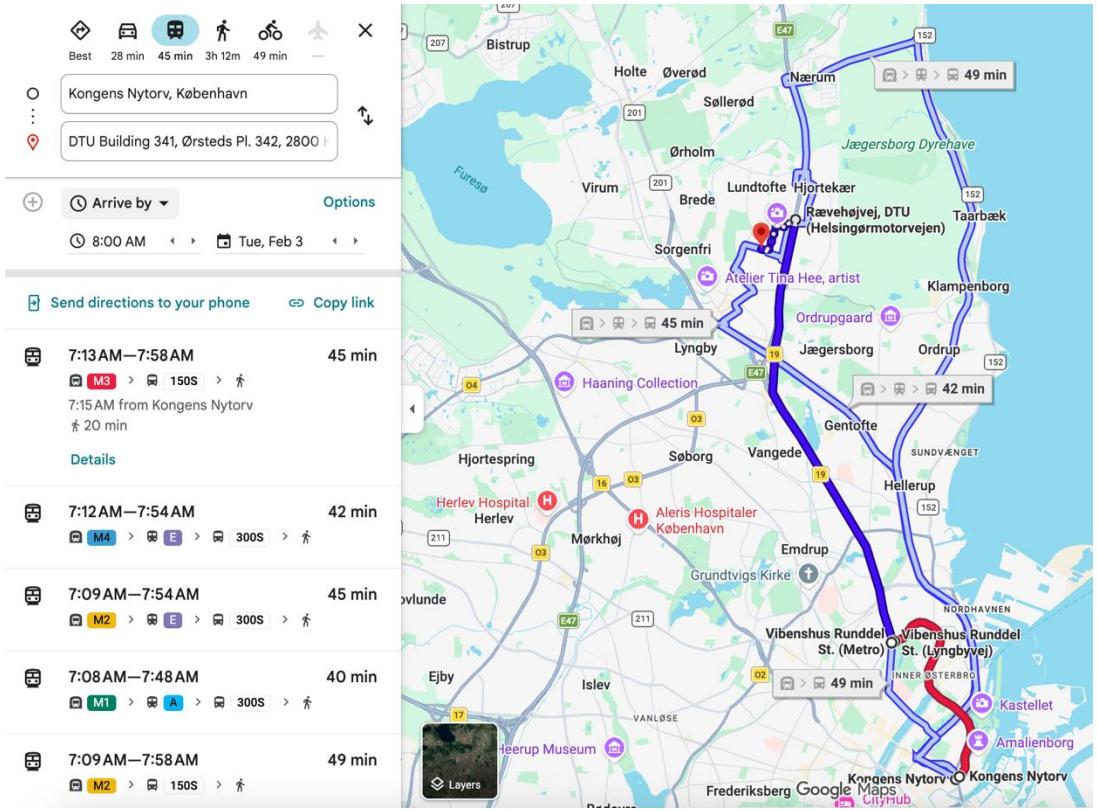
- Weather forecast



Massive fluid dynamics models. Navier-Stokes equations to simulate atmospheric pressure, temperature, and moisture across a 3D grid of the Earth

Mathematical Modeling EVERYWHERE

- Weather forecast
- GPS
- Traffic Flow models



- ***GPS: a system of equations based on the time it takes for signals to travel from at least four satellites***
- ***Traffic: Stochastic Models and Graph Theory to find the shortest path***

Mathematical Modeling EVERYWHERE

- Weather forecast
- GPS
- Traffic Flow models
- Recommendation systems



- Users and items as vectors in a high-dimensional space
- Distance between your tastes and millions of others
- Model predicts what you'll want to hear next

Mathematical Modeling EVERYWHERE

- Weather forecast
- GPS
- Traffic Flow models
- Recommendation systems
- Collaborative Filtering



- Users and items as vectors in a high-dimensional space
- Distance between your tastes and millions of others
- Model predicts what you'll want to hear next

Mathematical Modeling EVERYWHERE

- Weather forecast
- GPS
- Traffic Flow models
- Recommendation systems
- Collaborative Filtering
- …
- …
- …
- …
- …
- …

What is a model?

- Conceptual models – the foundation for our understanding of the world and each other

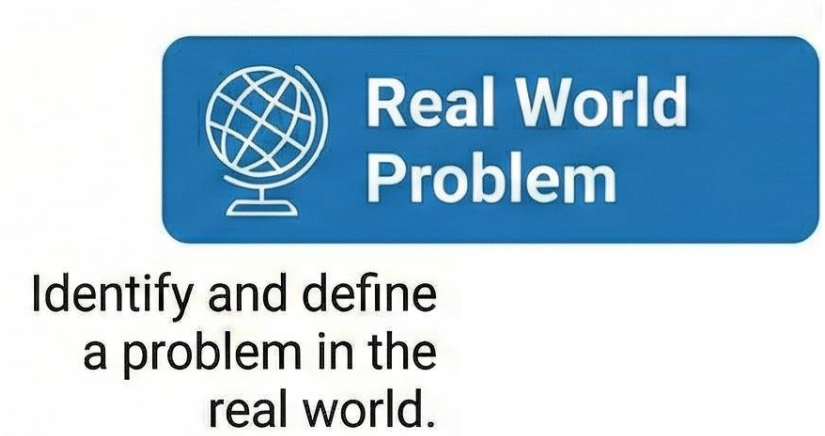
What is a model?

- Conceptual models – the foundation for our understanding of the world and each other
- *In vivo* models – in life. Lab rats, humans, living organisms
- *In vitro* models – in glass. Processes and organisms in petri dishes and test tubes
- *In situ* – in place. Testing the phenomenon where it occurs.
- In silico models– in silicon. In a computer program

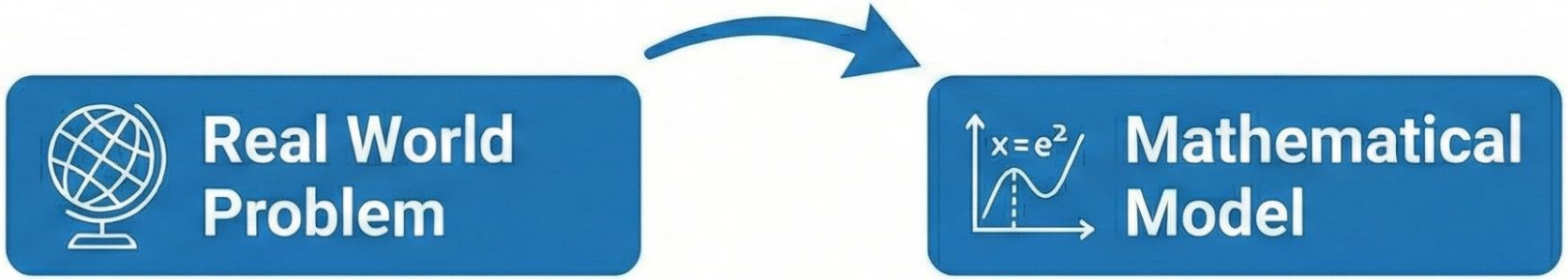
What is a model?

- Conceptual models – the foundation for our understanding of the world and each other
- *In vivo* models – in life. Lab rats, humans, living organisms
- *In vitro* models – in glass. Processes and organisms in petri dishes and test tubes
- *In situ* – in place. Testing the phenomenon where it occurs.
- In silico models– in silicon. In a computer program
- Mathematical models
  - Describe phenomena with numbers
  - Model relations
  - Computations are often necessary to find a solution
  - Theoretical models – basic phenomena e.g. in physics and chemistry
  - Empirical models – phenomena that are too complex to formulate all the theoretical connections
- Plenty of keywords to subcategories: Deterministic vs. stochastic, statistical, linear vs. non-linear, convex etc.

The modeling cycle



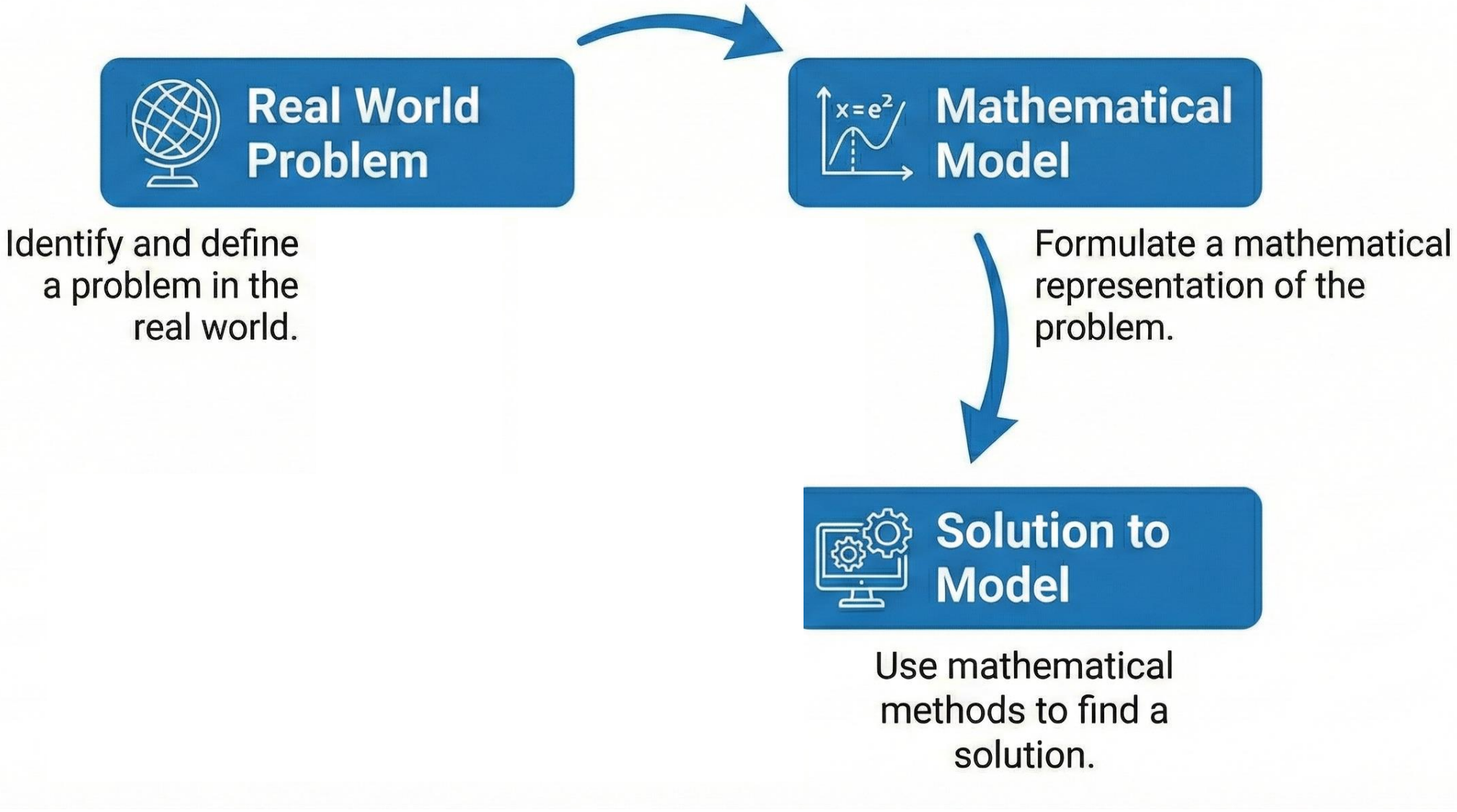
The modeling cycle



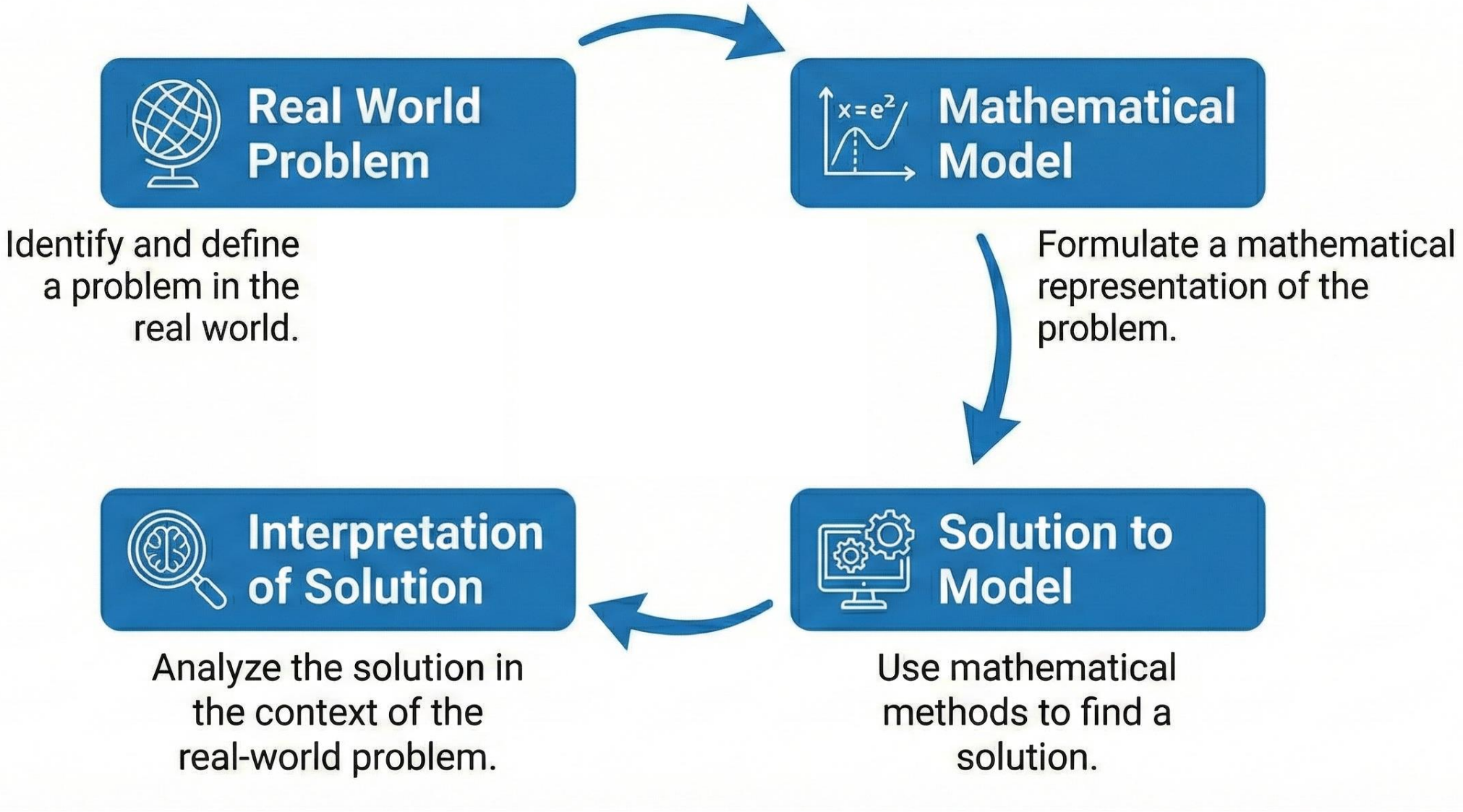
Identify and define a problem in the real world.

Formulate a mathematical representation of the problem.

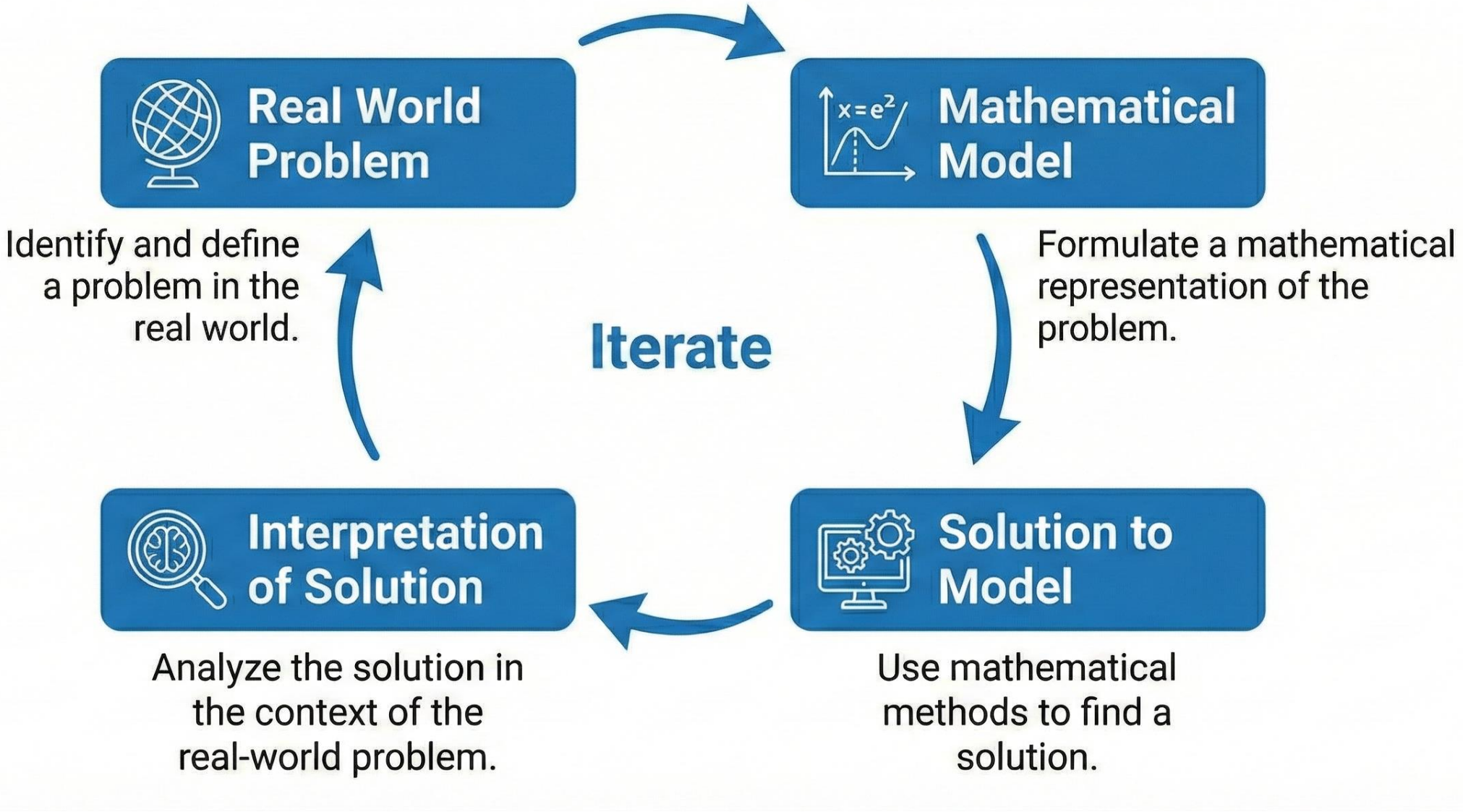
The modeling cycle



The modeling cycle



The modeling cycle



Commonalities in a model?



- **Exercise:** Establish a common concrete (but arbitrary) model as the basis.
- **Scenario set-up:** When travelling through Denmark, you have two options
  - **Bridge** or **ferry**
- **Modelling goal:**
  - Is taking the ferry the green (environmentally friendly) choice?

$$CO_2_boat < CO_2_driving$$

Brainstorming factors and parameters

– What affects either side of the model relation?

Join at menti.com | use code **5494 8990**



$CO_2_boat < CO_2_driving$

- Time: 5? Minutes
- Type in up to 3 words (factors / parameters) to the word cloud

- **Rules:**
- No capital letters
- Use **underscore** to join words, e.g. ‘mass_boat’
- Keywords: ‘boat’, ‘vehicle’, ‘passenger’, ‘engine’

Data

- Empirical observations
- Deterministic or stochastic
- **Deterministic data**
  - Clear relation between input and output
  - Dependent and independent variable
- **Stochastic data**
  - Random relation between input and output
  - Modelled by their distribution

Data

- **Deterministic data**
  - Movement of the planets
- **Stochastic data**
  - Stock markets

All models are wrong

“All models are wrong, but some are useful”

- George Box, British statistician

Introduction to exercise 1

Salami Quality

Why are we modeling Salami?



Why are we modeling Salami?



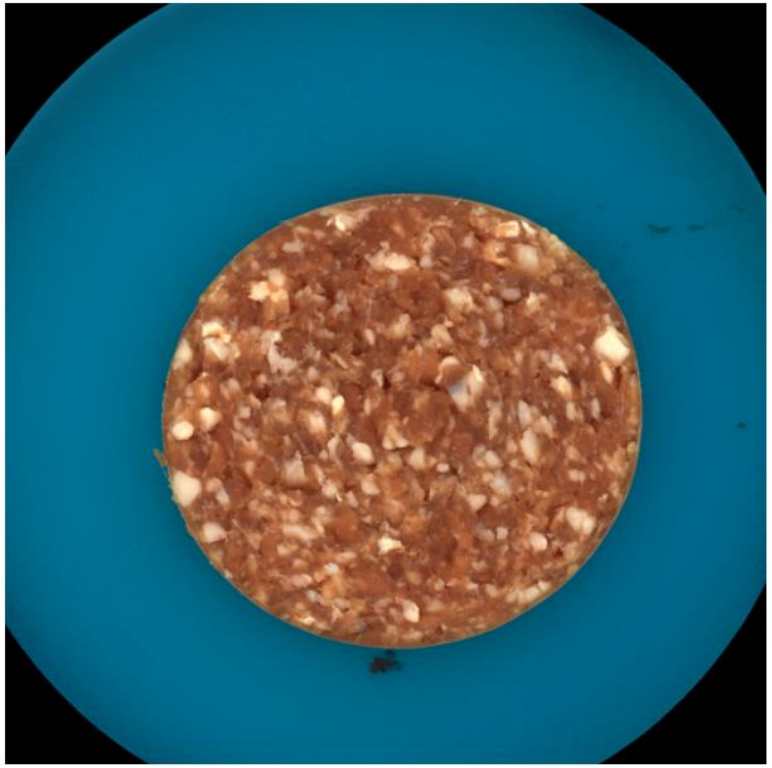
Food Safety is Critical

Poor quality control in cured meats can be fatal.
We are dealing with:

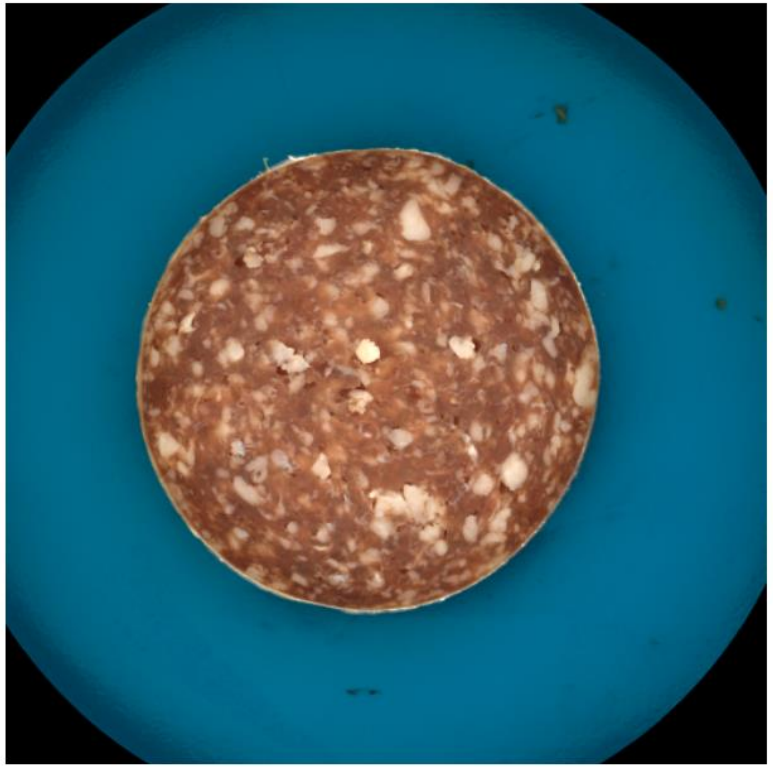
- **Listeria monocytogenes:** Mortality rate of 20-30%.
- **Clostridium botulinum:** Causes Botulism (German: *Wurstvergiftung*). One of the most lethal toxins known (LD_{50} of 1.3-2.1 ng/kg).

Our Goal: Develop a mathematical model (automated, non-invasive method) to assess the quality of a salami

The data



Day 1



Day 6



Day 28

The data: multispectral images

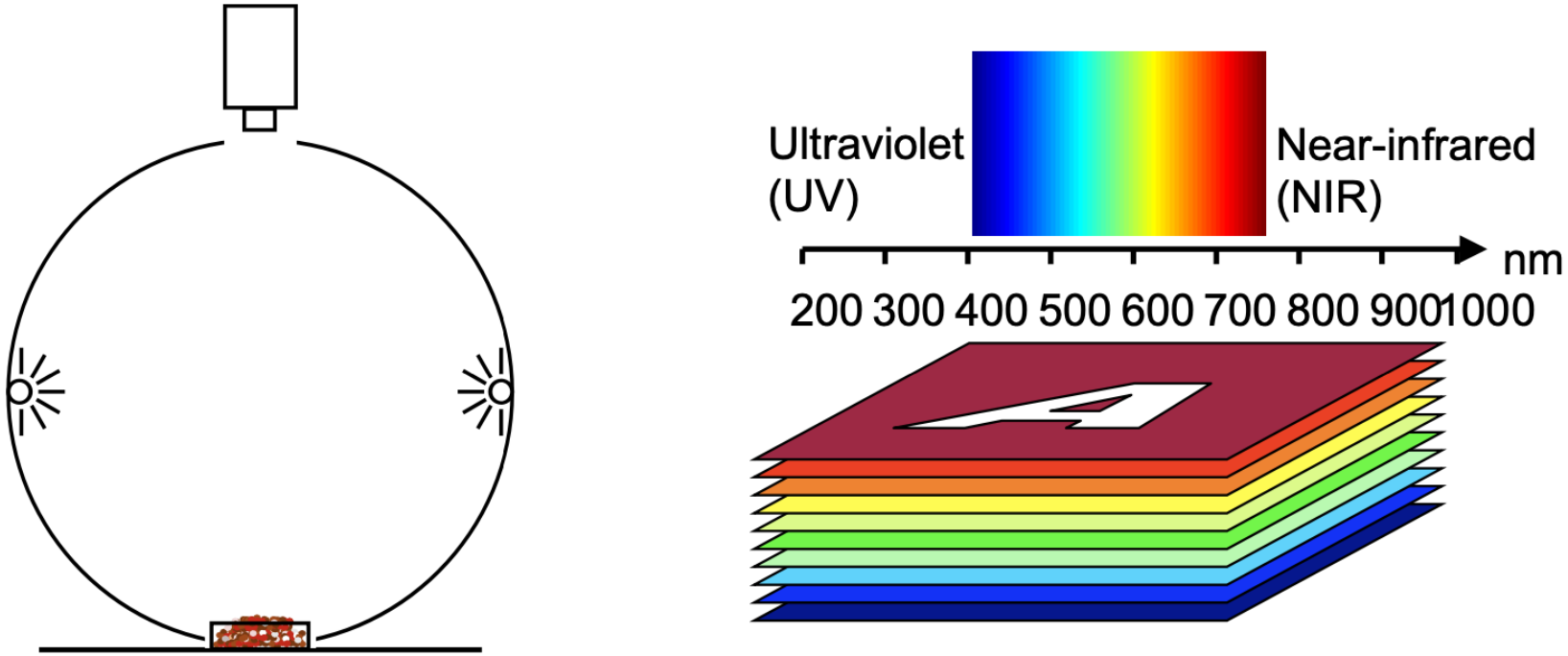
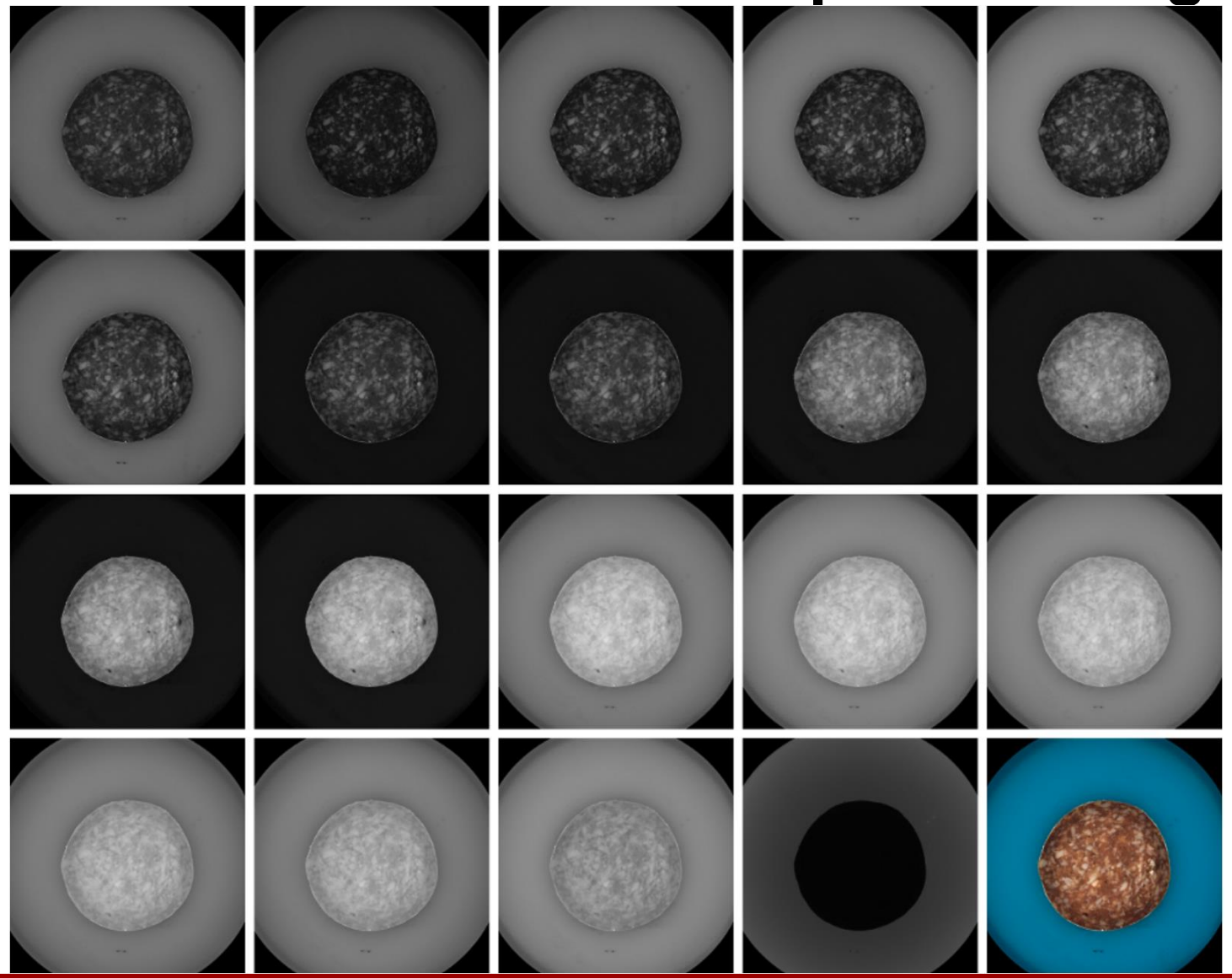


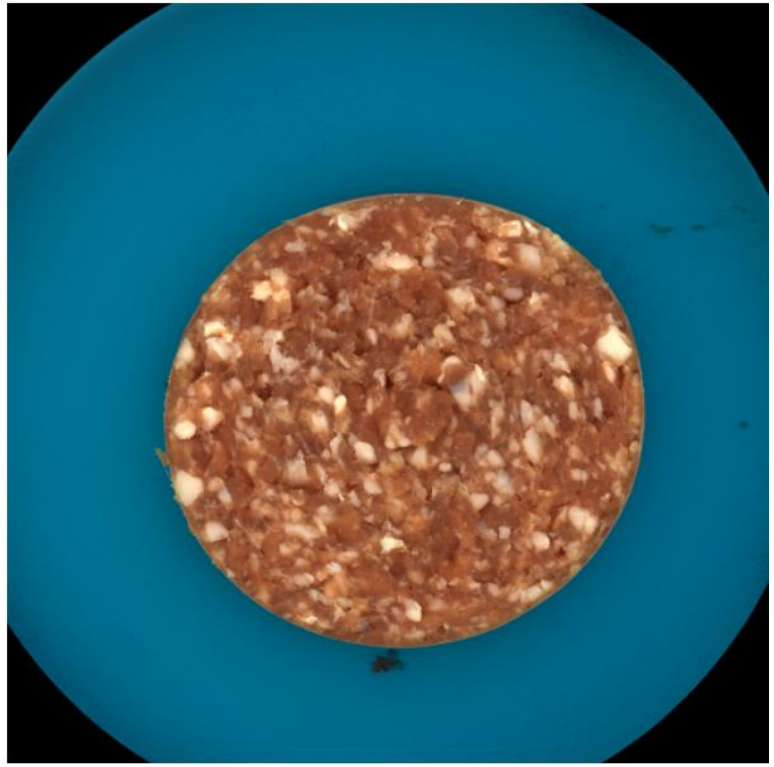
Figure 2: Sketch showing the principals of VideometerLab (left) and spectral images (right). VideometerLab is a sphere lit by coloured (specific wavelength) LEDs placed around the equator of the sphere. The resulting multi spectral image is a stack of images, where each layer is of a specific wavelength. The wavelengths range from blue to infra-red light.

The data: multispectral images



- Data from Dupont A/S (former Danisco A/S)
- A **batch** of salamis have been produced
- A number of salamis have been selected at day **1, 2, 3, 6, 13, 20 and 28**
- A slice of each salami has been imaged with **VideometerLab**
- We have a small subset of the images for the exercis

Salami Quality

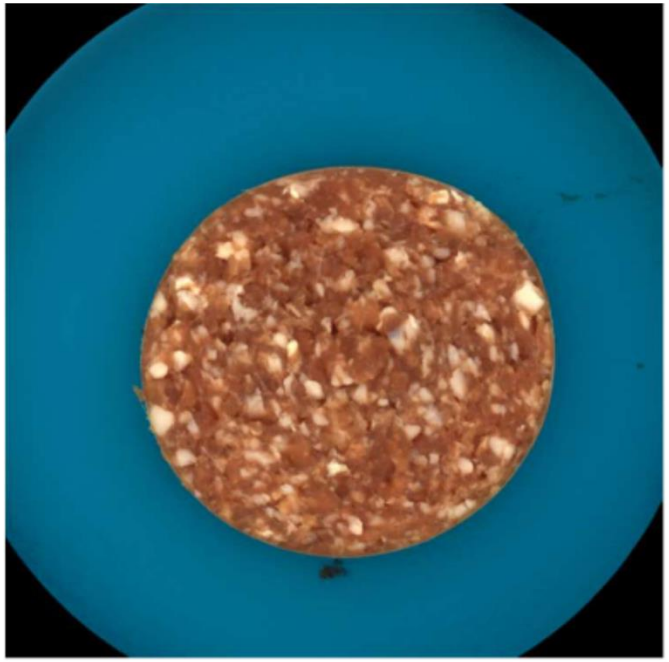


- **Quality parameters**
  - Contamination
  - Colour
  - Smell
  - Taste
  - Drying
  - **Meat and fat content**
  - ... and more!

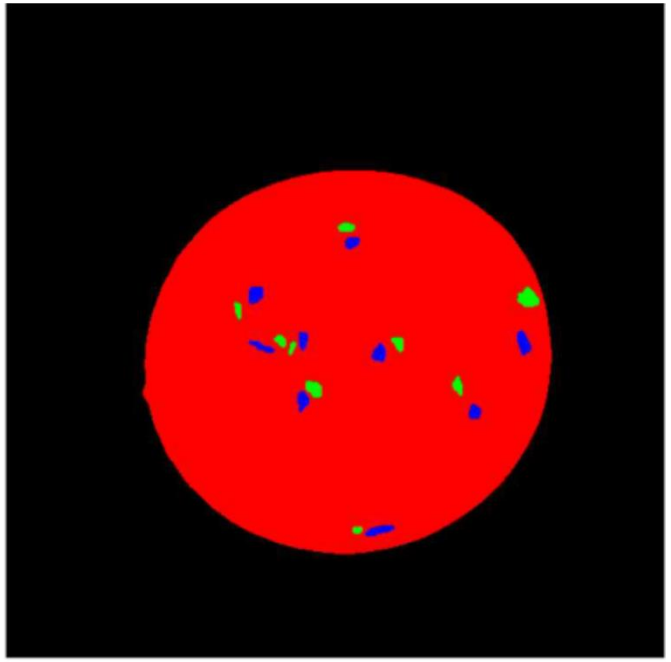
- We will look at the meat and fat content

Mathematical model

- **Problem**: Classify pixels to meat/fat
- **Training data**: known labels for pixels
- **Classifcation**: Supervised machine learning algorithm



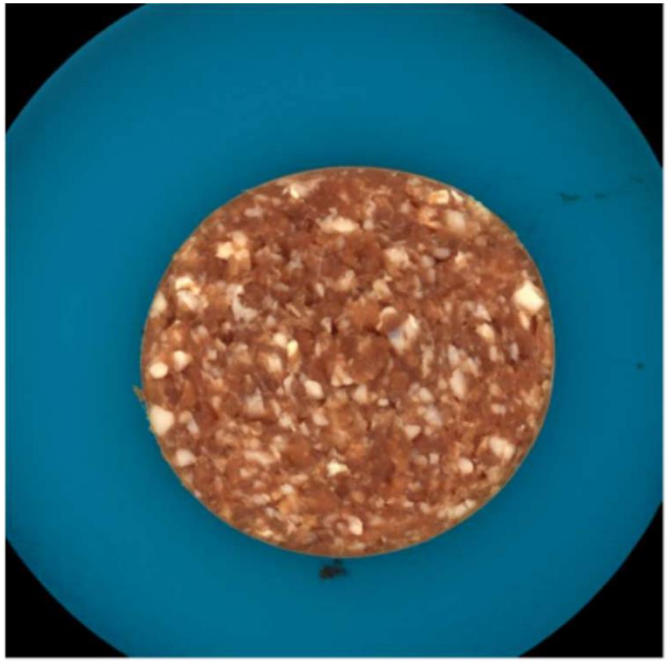
Day 1



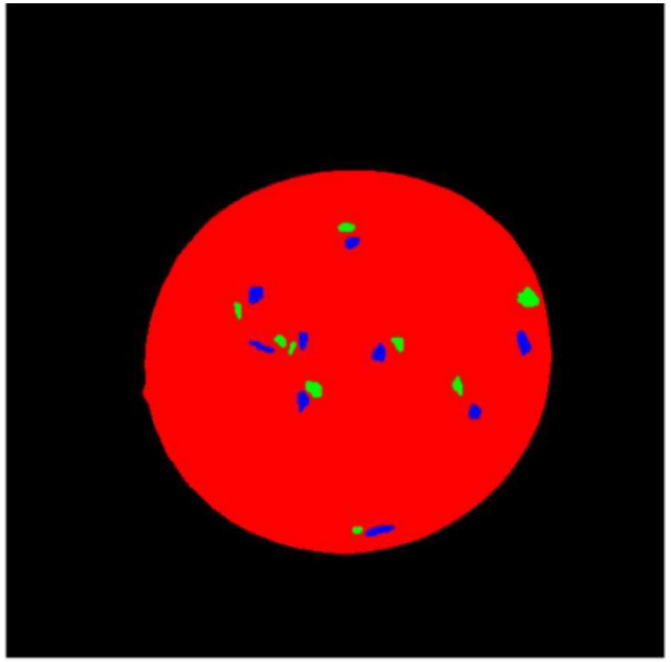
Annotation

Mathematical model

- **Problem**: Classify pixels to meat/fat
- **Training data**: known labels for pixels
- **Classifcation**: Supervised machine learning algorithm



Day 1



Annotation

Mathematical model (Warm-up, 1D Analysis)

Mathematical model (Warm-up, 1D Analysis)

Salmon

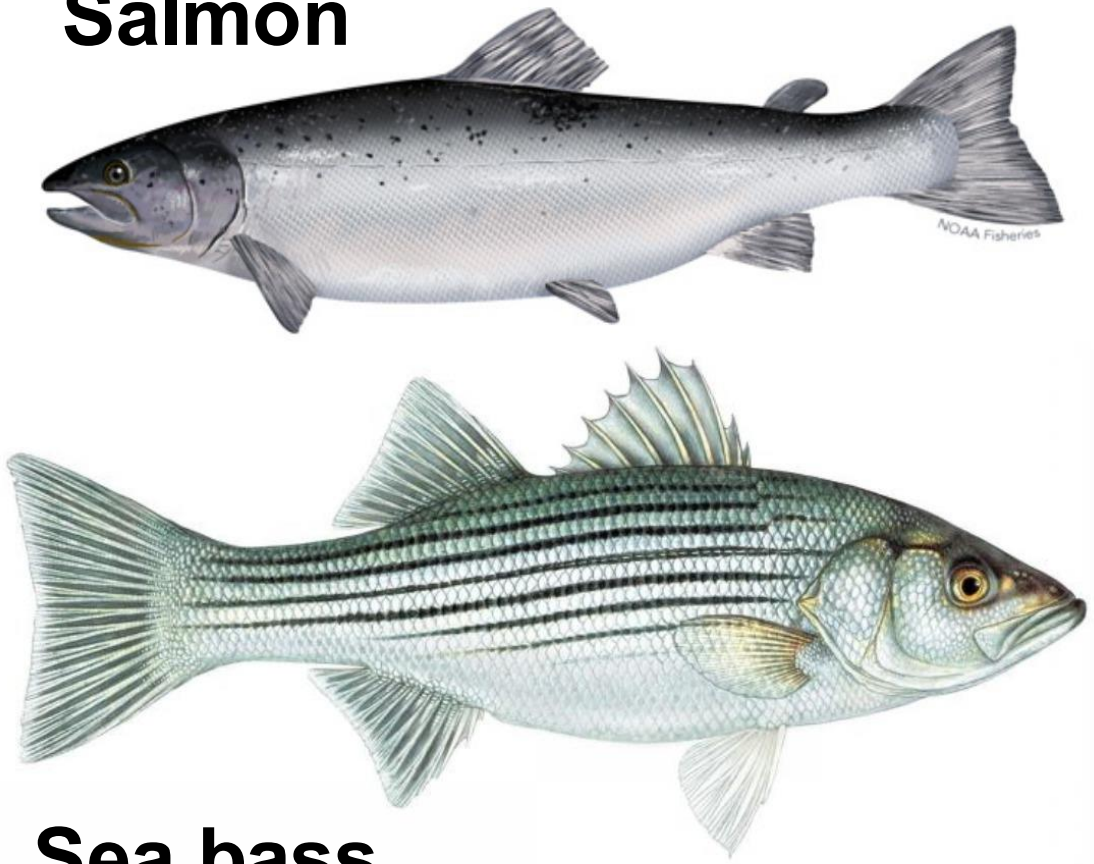




Sea bass

Mathematical model (Warm-up, 1D Analysis)

Salmon

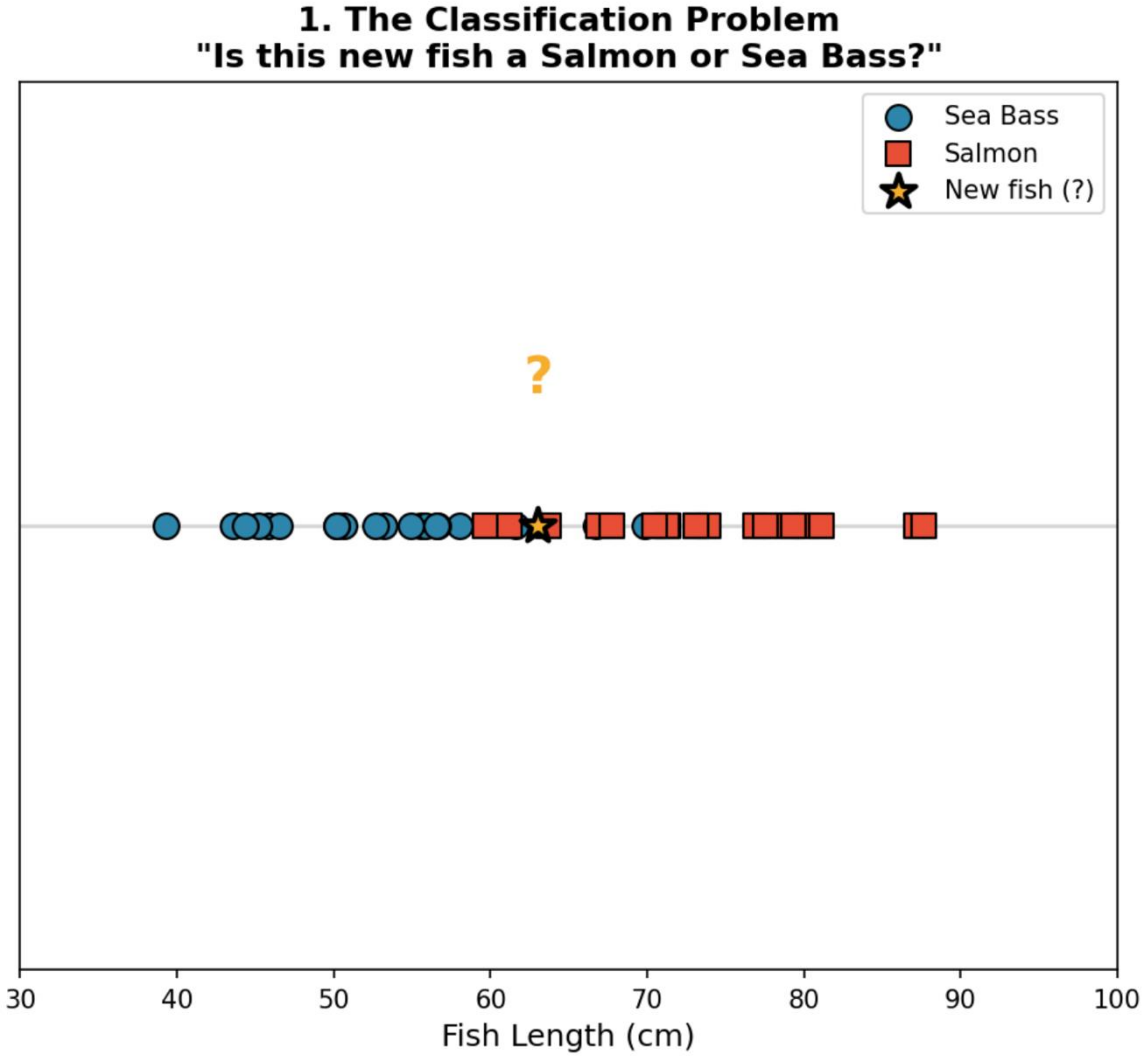


Sea bass

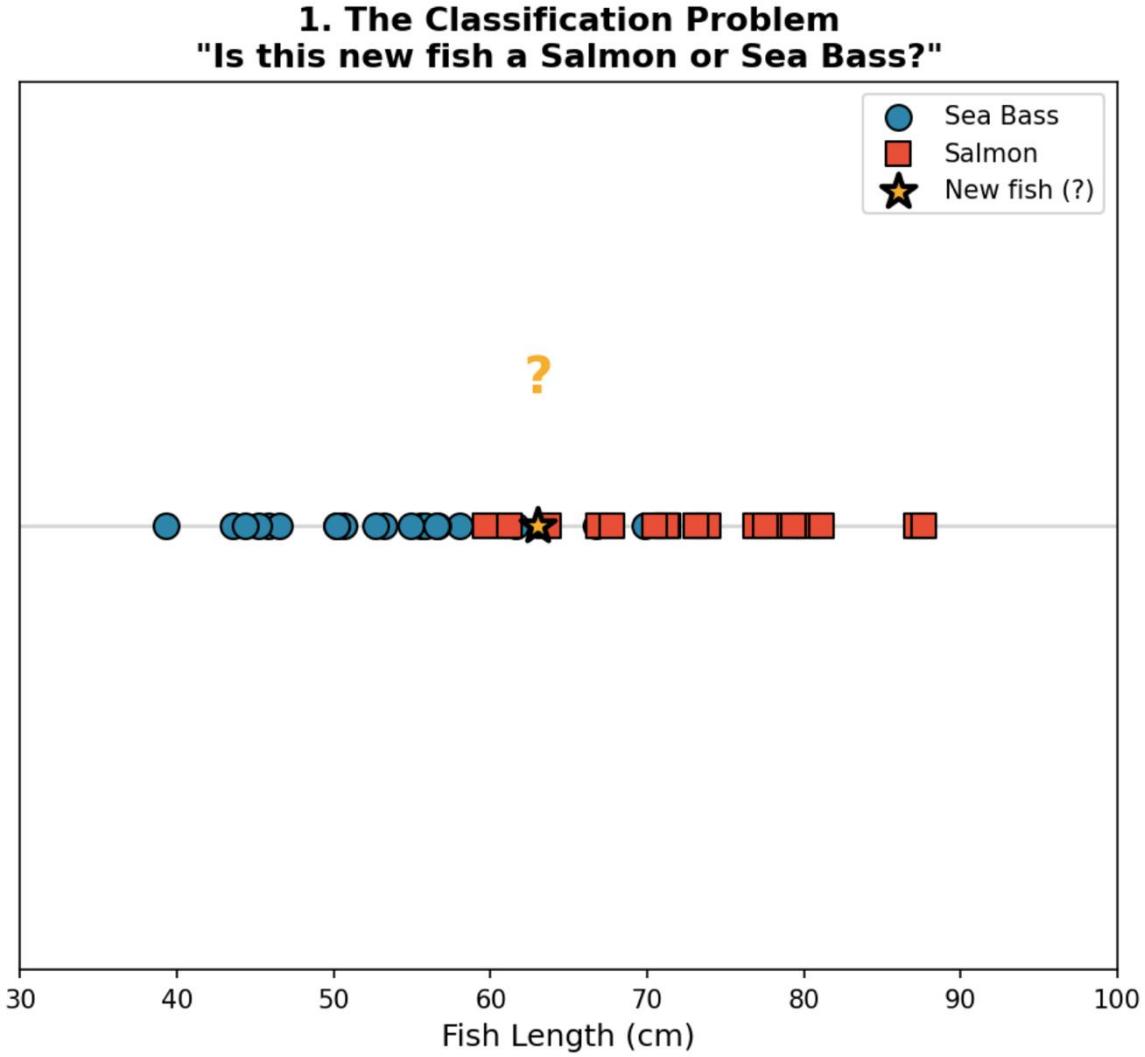
Training Data (Length in cm):

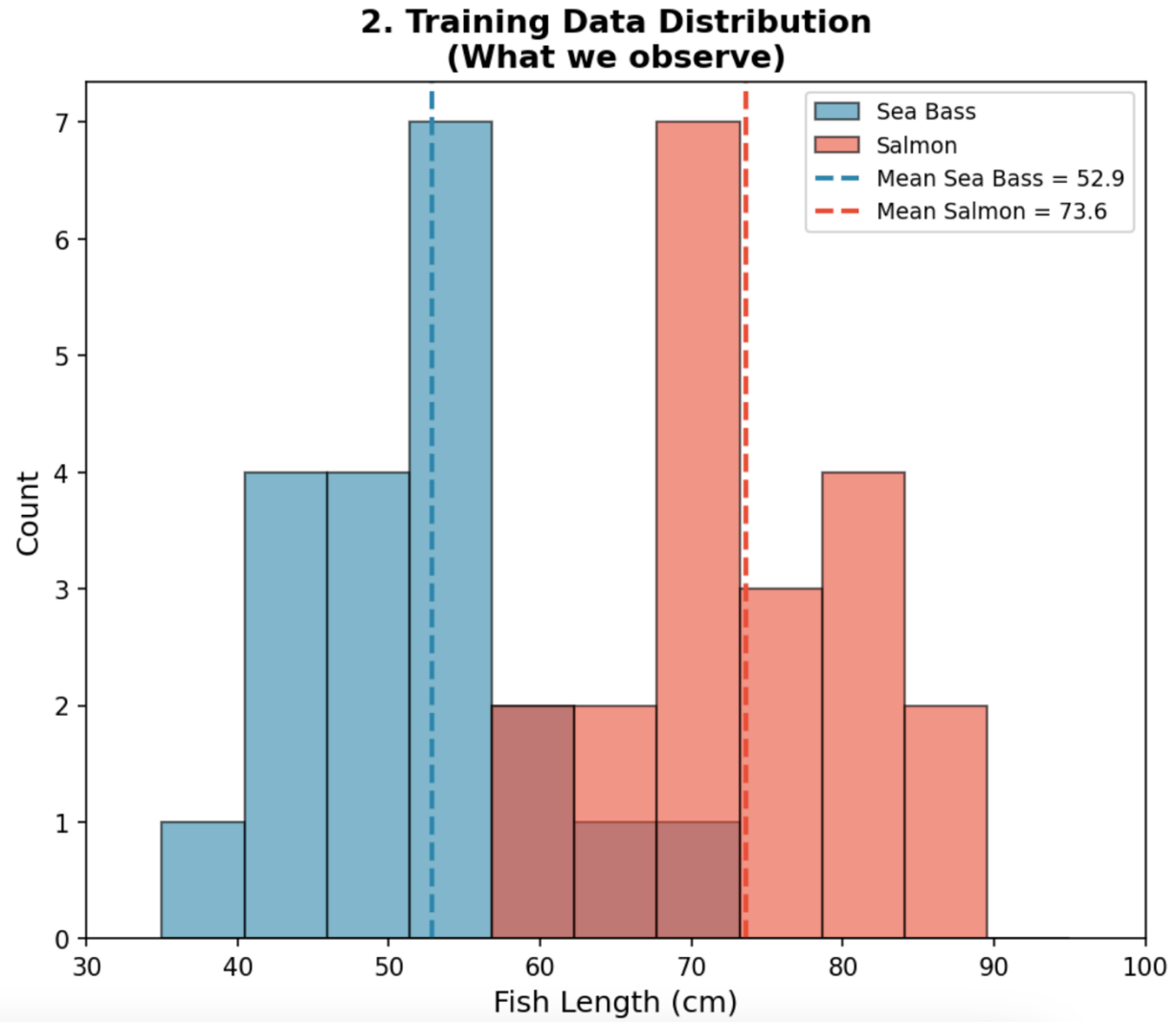
- Salmon: [79.0, 73.9, 80.2, 87.2, 73.1, 73.1, 87.6, 81.1, 71.2, 79.3, 71.3, 71.3, 76.9, 59.7, 61.2, 70.5, 66.9, 77.5, 67.7, 63.7]
- Sea Bass: [66.7, 53.2, 55.5, 43.6, 50.6, 55.9, 45.8, 58.0, 50.2, 52.7, 50.2, 69.8, 54.9, 46.5, 61.6, 45.2, 56.7, 39.3, 44.4, 56.6]

Linear Discriminant Analysis

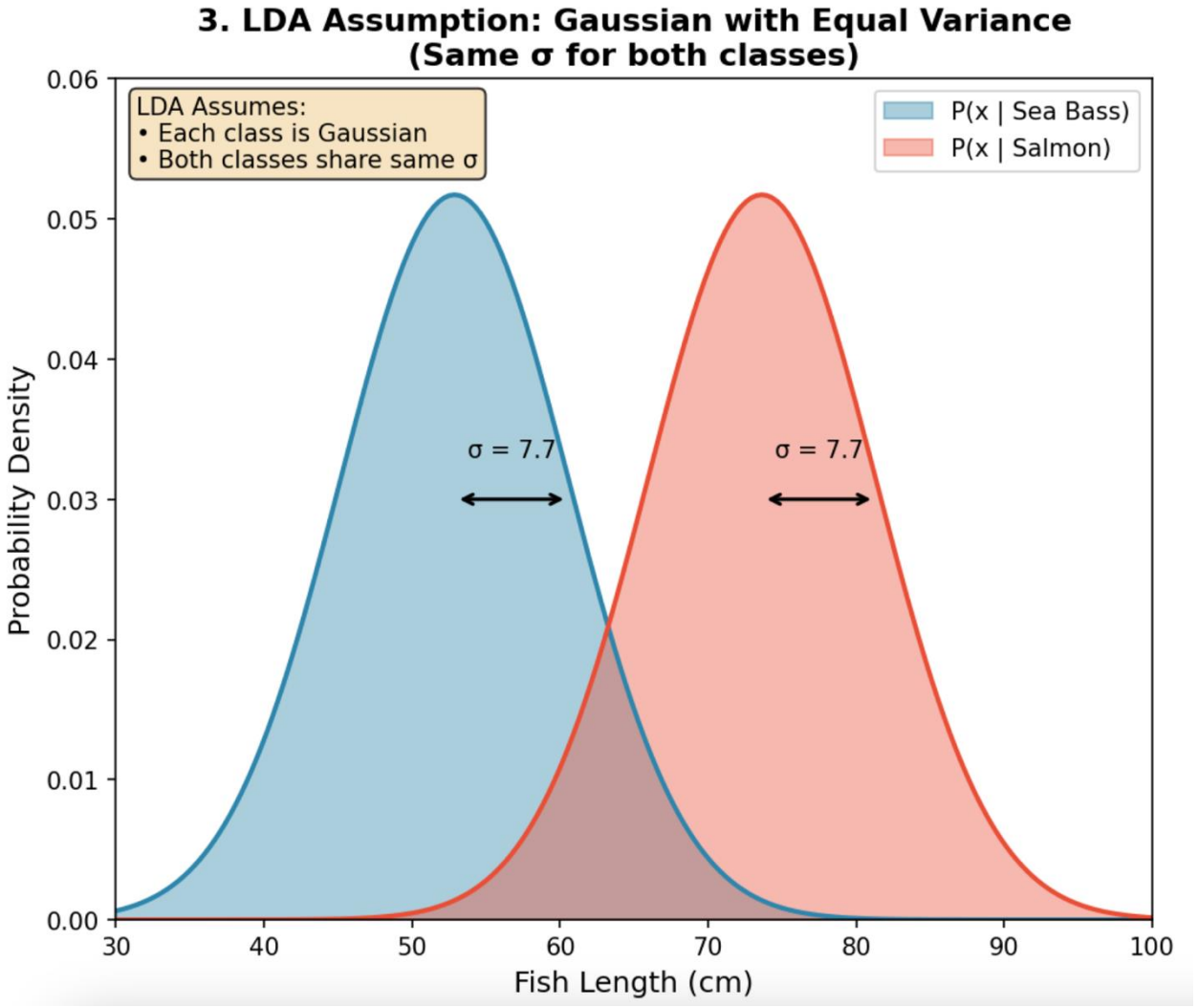


Linear Discriminant Analysis

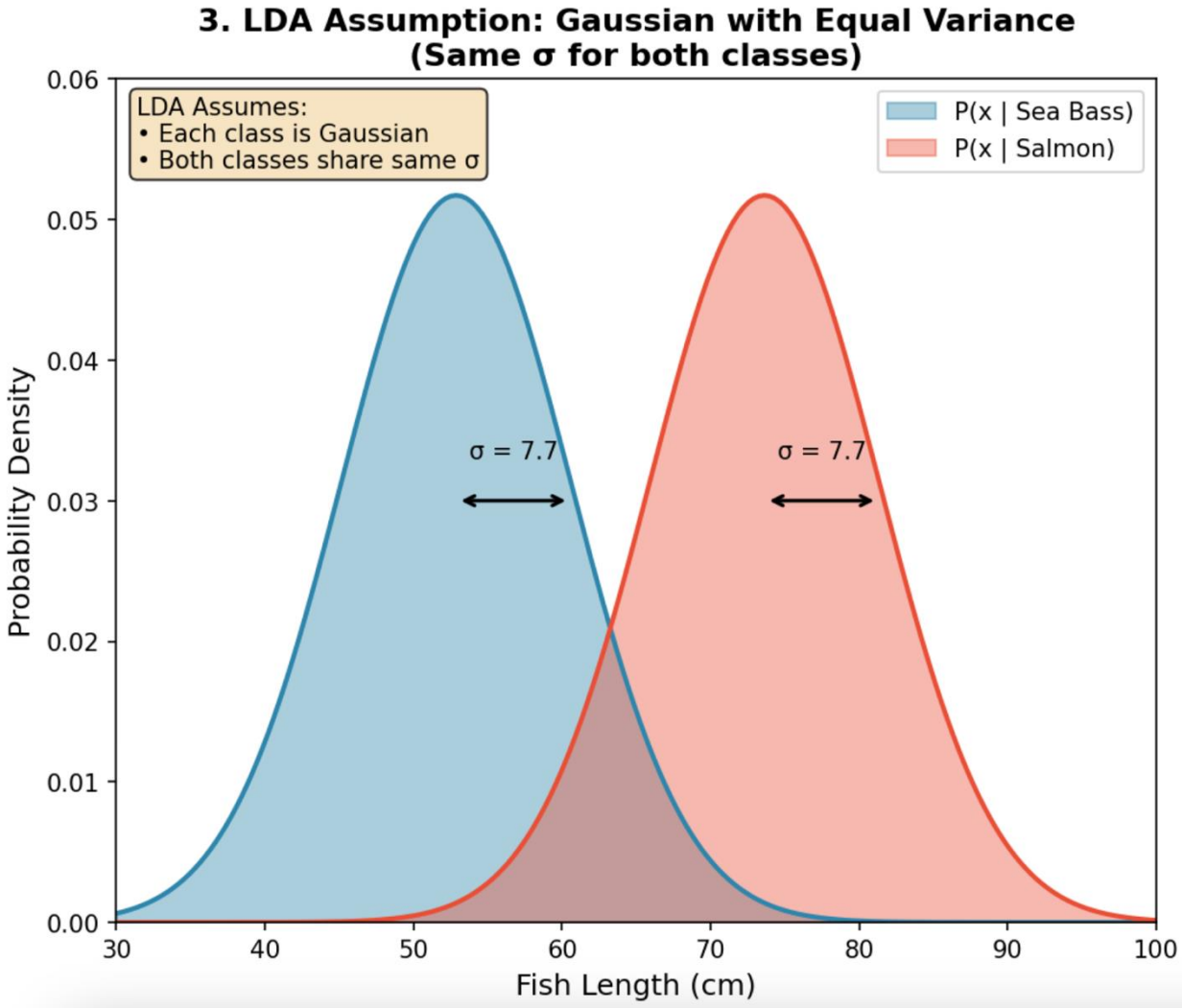


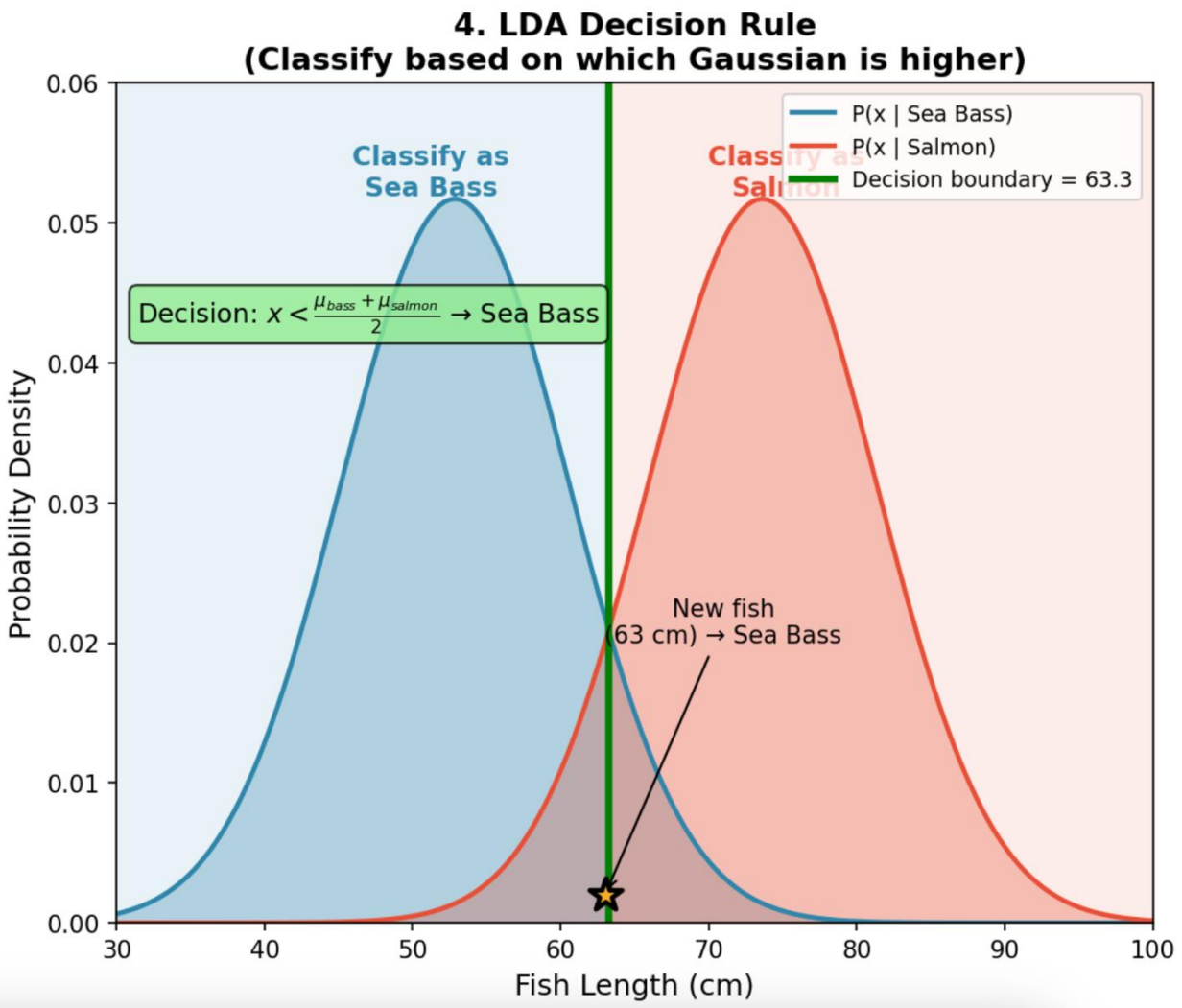


Linear Discriminant Analysis

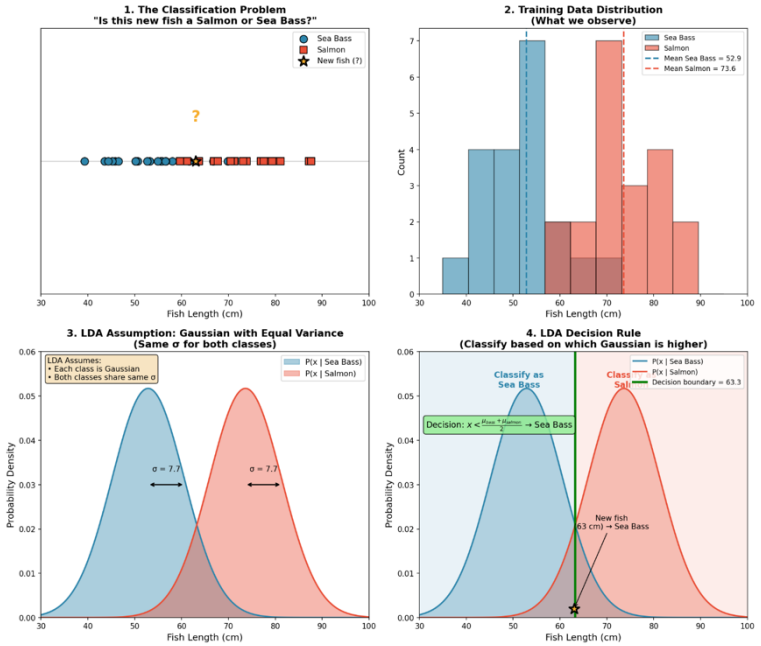


Linear Discriminant Analysis





Linear Discriminant Analysis



- **Independence of Observations**
  The data points are assumed to be sampled independently
- **Multivariate Normality**
  The independent variables (predictors) for each class are assumed to follow a normal (Gaussian) distribution
- **Homogeneity of Variance-Covariance**
  LDA assumes that all classes share the same variance-covariance matrix
- **Linearity**
  The relationship between the predictor variables is assumed to be linear

Problem and Assumptions

Linear Discriminant Analysis

Sea Bass: $\mathcal{D}_1 = \{x_1^{(1)}, x_2^{(1)}, \ldots, x_{n_1}^{(1)}\}$ where $n_1 = 20$

Salmon: $\mathcal{D}_2 = \{x_1^{(2)}, x_2^{(2)}, \ldots, x_{n_2}^{(2)}\}$ where $n_2 = 20$

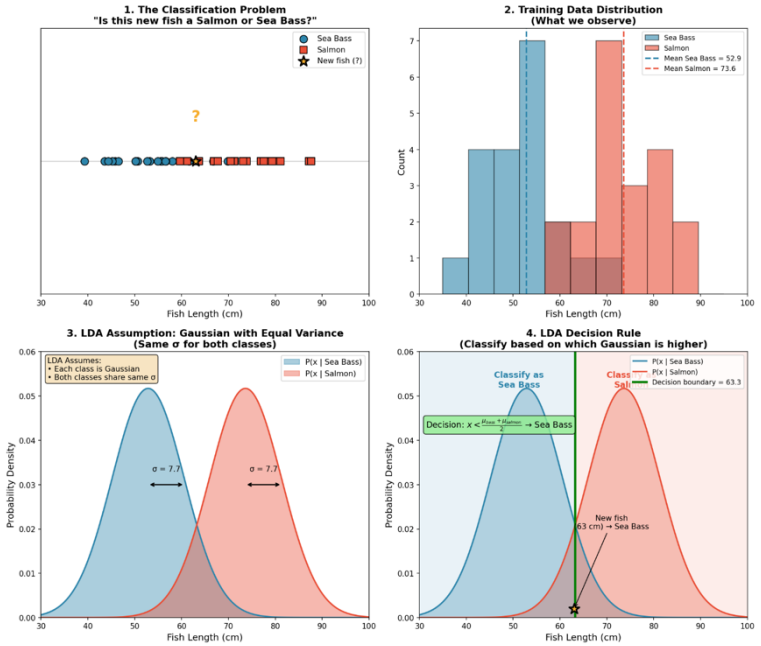
find $P(\text{class} \mid x^*)$

$$P(x \mid \text{Sea Bass}) = \mathcal{N}(x \mid \mu_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right)$$

$$P(x \mid \text{Salmon}) = \mathcal{N}(x \mid \mu_2, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right)$$

Key assumption: Both classes share the same σ^2 (homoscedasticity)

Prior probabilities: $\pi_1 = P(\text{Sea Bass})$ and $\pi_2 = P(\text{Salmon})$, with $\pi_1 + \pi_2 = 1$



Linear Discriminant Analysis

Estimate Parameters from Problem and Assumptions training data

Sea Bass: $\mathcal{D}_1 = \{x_1^{(1)}, x_2^{(1)}, \ldots, x_{n_1}^{(1)}\}$ where $n_1 = 20$

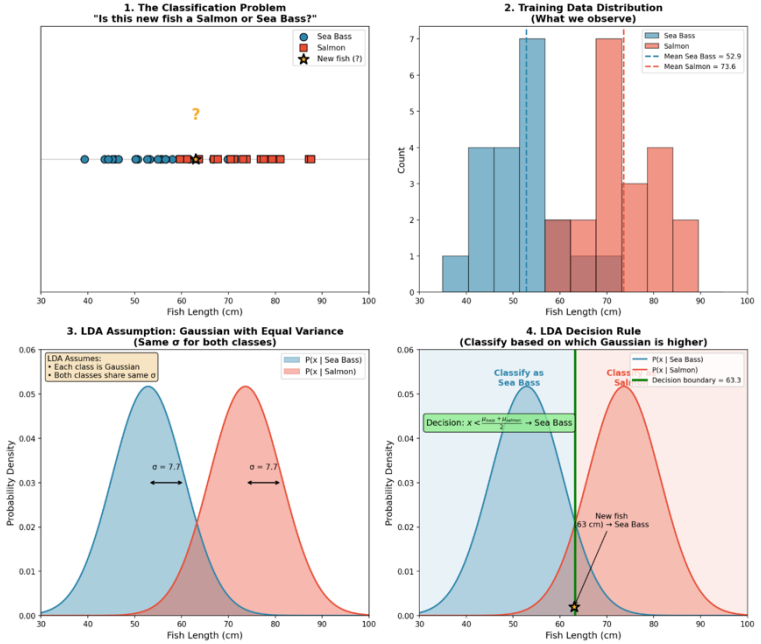
Salmon: $\mathcal{D}_2 = \{x_1^{(2)}, x_2^{(2)}, \ldots, x_{n_2}^{(2)}\}$ where $n_2 = 20$

find $P(\text{class} \mid x^*)$

$$P(x \mid \text{Sea Bass}) = \mathcal{N}(x \mid \mu_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right)$$

$$P(x \mid \text{Salmon}) = \mathcal{N}(x \mid \mu_2, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right)$$

Key assumption: Both classes share the same σ^2 (homoscedasticity)



Prior probabilities: $\pi_1 = P(\text{Sea Bass})$ and $\pi_2 = P(\text{Salmon})$, with $\pi_1 + \pi_2 = 1$

Class means (Maximum Likelihood Estimates):

$$\hat{\mu}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i^{(1)} = 52.87 \text{ cm} \quad \text{(Sea Bass)}$$

$$\hat{\mu}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} x_i^{(2)} = 73.63 \text{ cm} \quad \text{(Salmon)}$$

Pooled variance estimate (using both classes):

$$\hat{\sigma}^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2} = \frac{19 \cdot s_1^2 + 19 \cdot s_2^2}{38}$$

$$\hat{\sigma} = 7.71 \text{ cm}$$

Linear Discriminant Analysis

Apply Bayes' Rule

Posterior probability using Bayes' theorem:

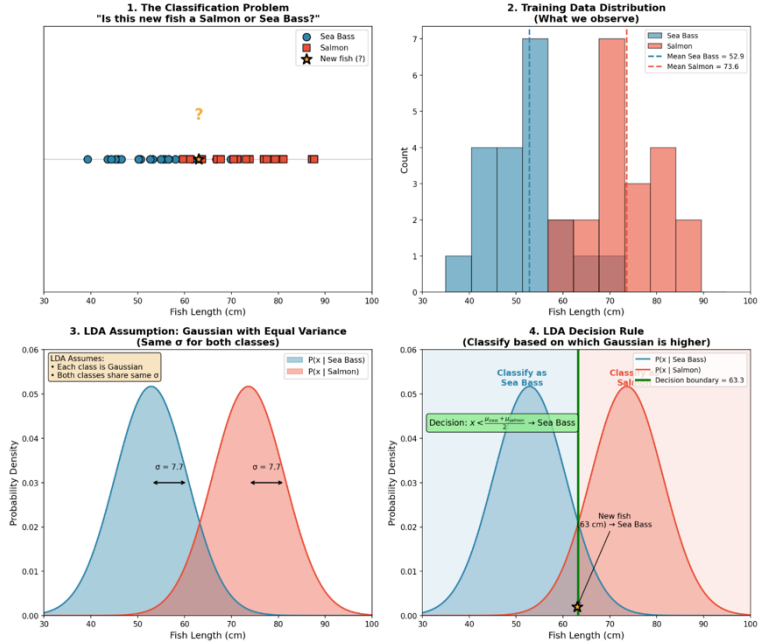
$$P(\text{class}_k \mid x) = \frac{P(x \mid \text{class}_k) \cdot \pi_k}{P(x)} = \frac{P(x \mid \text{class}_k) \cdot \pi_k}{\sum_j P(x \mid \text{class}_j) \cdot \pi_j}$$

Decision rule: Classify to the class with higher posterior. Equivalently, compare:

$$P(x \mid \text{Sea Bass}) \cdot \pi_1 \quad \text{vs} \quad P(x \mid \text{Salmon}) \cdot \pi_2$$

Taking the log ratio (log-odds) and simplifying:

$$\log \frac{P(\text{Sea Bass} \mid x)}{P(\text{Salmon} \mid x)} = \log \frac{\pi_1}{\pi_2} + \frac{\mu_1 - \mu_2}{\sigma^2} x + \frac{\mu_2^2 - \mu_1^2}{2\sigma^2}$$



Linear Discriminant Analysis

Apply Bayes' Rule

Posterior probability using Bayes' theorem:

$$P(\text{class}_k \mid x) = \frac{P(x \mid \text{class}_k) \cdot \pi_k}{P(x)} = \frac{P(x \mid \text{class}_k) \cdot \pi_k}{\sum_j P(x \mid \text{class}_j) \cdot \pi_j}$$

Decision rule: Classify to the class with higher posterior. Equivalently, compare:

$$P(x \mid \text{Sea Bass}) \cdot \pi_1 \quad \text{vs} \quad P(x \mid \text{Salmon}) \cdot \pi_2$$

Find Decision Boundary

Taking the log ratio (log-odds) and simplifying:

$$\log \frac{P(\text{Sea Bass} \mid x)}{P(\text{Salmon} \mid x)} = \log \frac{\pi_1}{\pi_2} + \frac{\mu_1 - \mu_2}{\sigma^2} x + \frac{\mu_2^2 - \mu_1^2}{2\sigma^2}$$

With equal priors ($\pi_1 = \pi_2 = 0.5$), the decision boundary is where posteriors are equal:

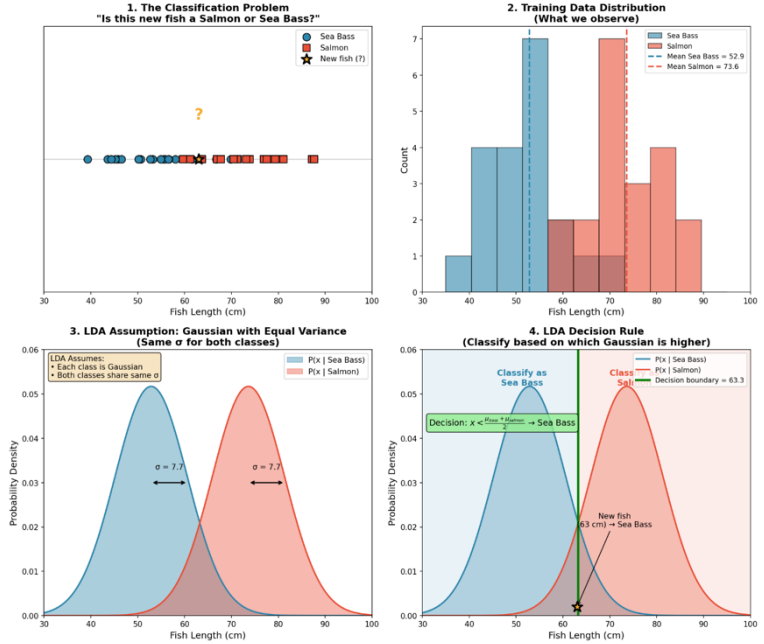
$$P(x \mid \text{Sea Bass}) = P(x \mid \text{Salmon})$$

Solving for x (the Gaussians intersect at):

$$x^* = \frac{\mu_1 + \mu_2}{2} = \frac{52.87 + 73.63}{2} = 63.25 \textbf{ cm}$$

LDA Decision Rule:

If $x < 63.25 \rightarrow$ classify as Sea Bass; If $x \geq 63.25 \rightarrow$ classify as Salmon



Multivariate Linear Discriminant Analysis

$$f_i(\mathbf{x}) = \frac{1}{\sqrt{2\pi}^n \sqrt{\det \hat{\mathbf{\Sigma}}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \hat{\boldsymbol{\mu}}_i)^T \hat{\mathbf{\Sigma}}^{-1}(\mathbf{x} - \hat{\boldsymbol{\mu}}_i)\right)$$

PDF of obsevavation **x**

$$S_i(\mathbf{x}) = \mathbf{x}^T \hat{\mathbf{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_i - \frac{1}{2} \hat{\boldsymbol{\mu}}_i{}^T \hat{\mathbf{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_i + \ln p_i$$

Discriminant function S

We classify a pixel **x** to class *i* if $S_i(x)$ is maximized

x: The 19-dimensional pixel vector

Σ: The pooled Covariance Matrix

μ: The mean spectral vector for class *i*

p_i: The prior probability

Experiments

- Implement the mathematical model – linear discriminant analysis
- Find the model parameters – day 1 is the training set
- Classify pixels in the training data – test if the model works at all
- Classify pixels in the entire salami for day 1
- Classify data for the remaining days
- Find the best day to train on
- Impact of prior knowledge

Analysis

- Quantitative analysis with regards to manual annotations
- Qualitative analysis by looking at the segmentation
- Consider - among others - the following
  - How good should the results be?
  - What are we comparing to?
  - How good is the model?
  - What could improve the measurement of meat and fat?

Report

Coherent report which independently describes the analysis

- Introduction with problem statement
- Description of method – the mathematical model (general)
- Description of experiment, analysis, and results (concrete)
- Discussion of results:
  - Does the method solve the problem?
  - How good is the solution?
  - What parameters have been used?
  - Are the data representative of the problem?
  - What are the limitations of the method?
- Conclusion – what is the contribution of the analysis

Questions???

1) Feedback
2) Form your group
3) Time for the exercise!

Join at menti.com | use code 7356 1361

